

How to Extend The Semantic Tableaux And Cut-Free Versions of the Second Incompleteness Theorem Almost to Robinson's Arithmetic Q

Dan E. Willard *

This paper was published by the Journal of Symbolic Logic 67 (2002) pp. 465-496.

Abstract

Let us recall that Raphael Robinson's Arithmetic Q is an axiom system that differs from Peano Arithmetic essentially by containing no Induction axioms [13, 18]. We will generalize the semantic-tableaux version of the Second Incompleteness Theorem *almost to the level* of System Q. We will prove that there exists a single rather long Π_1 sentence, valid in the standard model of the Natural Numbers and denoted as V , such that if α is *any* finite consistent extension of $Q + V$ then α will be unable to prove its Semantic Tableaux consistency. The same result will also apply to axiom systems α with infinite cardinality when these infinite-sized axiom systems satisfy a minor additional constraint, called the *Conventional Encoding Property*.

Our formalism will also imply that the semantic-tableaux version of the Second Incompleteness Theorem generalizes for the axiom system $I\Sigma_0$, as well as for all its natural extensions. (This answers an open question raised twenty years ago by Paris and Wilkie [15].)

1 INTRODUCTION

As originally formulated by Gödel, the Second Incompleteness Theorem discussed the inability of any extension of Peano Arithmetic to verify its own consistency when its proofs were constructed using a deductive calculi similar to a Hilbert (or Frege) type formalism. When Smullyan introduced his Semantic Tableaux version [20] of Gentzen's cut-free Sequent Calculus, it was realized that all extensions of Peano Arithmetic would also be unable to verify their own consistency if the Semantic Tableaux trees replaced the Frege-Hilbert methodologies as the underlying deductive calculi for generating the formal proofs.

But what about axiom systems weaker than Peano Arithmetic? *Would they also be inherently unable to verify their own consistency under the Hilbert and Semantic Tableaux styles of deductive calculi?*

Some initial research into this question was done by Bezboruah and Shepherdson [5], but it required a surprising 54 years after the publication of Gödel's Incompleteness Theorem for mathematicians to begin to formulate a full answer to it. Let us recall that Robinson's Arithmetic Q differs from Peano Arithmetic by containing no Induction axioms [13, 18]. In 1985, Pudlák proved [16] that no conceivable extension α of Q could formally verify its own consistency when a Hilbert-style form of deductive calculi was used to generate the formal proof structures. Moreover, Robert Solovay (private communications, [21]) combined several formalisms of Pudlák, Nelson [14] and Wilkie-Paris [24] to observe that Pudlák's

*Computer Science Dept, SUNYA, Albany, NY 12222 or dew@cs.albany.edu. Supported by NSF Grant CCR 99-02726

theorem generalized to systems weaker than Q in that they would recognize Successor (but not Addition and Multiplication) as total functions.

The main gap left by the prior literature was that it had not resolved fully whether or not the Second Incompleteness Theorem was also valid for weak systems with respect to Cut-Free forms of deductive calculi, such as Semantic Tableaux. For instance in 1981, Paris and Wilkie [15] raised the open question whether IS_0 or other axioms very near it could verify their own Cut-Free Consistency?

More specifically, let us say an axiom system α has an ability to verify its **tableaux-based consistency** iff α can formally prove the non-existence of a proof of $0=1$ in a deduction system that uses **simultaneously** α as the set of proper axioms and Smullyan's rules for Semantic Tableaux [7, 20] as the employed deductive calculi. In this context, the system IS(A) from [25, 26] illustrated one method for evading the force of the Second Incompleteness Theorem. IS(A) used an atomic symbol $M(x, y, z)$ to represent that the multiplicative product of x times y equals z , and it did not recognize Multiplication as a total function satisfying the usual identity " $\forall x \forall y \exists z : M(x, y, z)$ ". IS(A)'s example showed that some axiom systems can simultaneously:

1. recognize their "tableaux-based" consistency (in the formal sense defined above),
2. recognize Addition as a total function, and
3. prove all the Π_1 -like theorems of Peano Arithmetic in a slightly modified language that replaces the Multiplication Function symbol with a $M(x, y, z)$ -atomic predicate symbol.

Moreover, there are other types of semantic tableaux exceptions to the Second Incompleteness Theorem, besides those mentioned in the paragraph above. To explain the nature of these other semantic-tableaux caveats to the Second Incompleteness Theorem, we must remind ourselves that a formula $\Upsilon(v)$ is said to be a **Definable Cut** for an axiom system α if α can formally prove:

$$\Upsilon(0) \text{ and } \forall v \{ \Upsilon(v) \Rightarrow \Upsilon(v+1) \} \quad (1)$$

The prior literature [11, 12, 14, 16, 22, 24] has illustrated several examples of different formulae $\Upsilon(v)$, which are Definable Cuts for α , such that α can prove its *Semantic Tableaux* consistency local to these Cuts. Thus if " $\text{SemPrf}_\alpha(x, y)$ " denotes that y is a semantic tableaux proof of the theorem x and if $\lceil \Psi \rceil$ designates Ψ 's Gödel number, the prior literature has shown how several quite different axiom systems α can prove their cut-localized consistency statement below:

$$\forall y \{ \Upsilon(y) \Rightarrow \neg \text{SemPrf}_\alpha(\lceil 0=1 \rceil, y) \} \quad (2)$$

For instance, Nelson [14] showed that Robinson's Arithmetic Q (with linear ordering) can formally prove a version of (2) about itself, where Υ is defined in Nelson's book. Pudlák [9, 16] proved a more general theorem showing a similar effect was applicable to any finitely axiomatized sequential theory (and also allowing for Wilkie-Paris's notion [24] of a Herbrand-restricted-consistency).

Another interesting aspect about Equation (2)'s partial evasion of the Second Incompleteness Theorem is that Pudlák proved that its analog *would fail* for Hilbert deduction. Thus, Equation (2) illustrates a second example where Semantic Tableaux Deduction generates many more caveats to the Second Incompleteness Theorem than does Hilbert-style deduction.

In light of these two examples about how the Second Incompleteness Theorem can be partially evaded by the semantic tableaux deductive calculi, it is noteworthy that Adamowicz and Zbierski [2, 3] have proven that the axiom system $I\Sigma_0 + \Omega_2$ satisfies the Second Incompleteness property for several forms of cut-free deduction. In particular, they provide several different proofs of forms of this Incompleteness theorem at the $I\Sigma_0 + \Omega_2$ level, and an unpublished paper by Adamowicz [1] also establishes a cut-free version of the Second Incompleteness Theorem for $I\Sigma_0 + \Omega_1$. In these papers, $I\Sigma_0$ refers to the frequently-studied axiom system [9, 24] that is weaker than Peano Arithmetic by recognizing the validity of Induction only for Σ_0 formulae, and Ω_i is defined in the usual standard manner, identical to that in say [9, 24]. (In particular, if $\omega_1(x)$ denotes the well-known function that maps an integer x onto $x^{\log x}$ and if $\omega_2(x)$ denotes the slightly faster growing function that maps analogously an integer x onto $2^{2^2 \log \log x}$, then the axiom Ω_i states essentially that $\omega_i(x)$ is a total function.)

The new versions of the Second Incompleteness Theorem by Adamowicz-Zbierski give rise to the following two natural questions:

1. Where between IS(A)'s evasion of the Second Incompleteness Theorem and $I\Sigma_0 + \Omega_1$'s and $I\Sigma_0 + \Omega_2$'s obeying of it, does the Second Incompleteness effect become valid for Semantic Tableaux deduction?
2. Does the Second Incompleteness Effect for Semantic Tableaux proofs become valid for Robinson's Arithmetic Q, or other axiom systems very near it?

We will offer a partial (albeit not complete) answer to these questions.

We will prove there exists a single rather long Π_1 sentence, valid in the standard model of the Natural Numbers and denoted as V , such that if α is *any* consistent and finite extension of $Q + V$ then α will be unable to verify its own Semantic Tableaux consistency. The same result will also apply to axiom systems α with infinite cardinality that satisfy a minor additional constraint, called the *Conventional Encoding Property* (which is defined in Section 5).

The main reason the above theorem is interesting is that the fastest growing function recognized by $Q + V$ is multiplication. In other words, axioms similar to Ω_1 (indicating that the operation $\omega_1(x)$ is a total function) will be unprovable from $Q + V$. The absence of the axiom Ω_1 is interesting because statements of at least its strength were previously needed to establish generalizations of the Second Incompleteness Theorem for Semantic Tableaux and other cut-free deductive calculi [1, 2, 3].

Moreover as has already been mentioned, if we chose to drop the axiom that Multiplication is a total function (and treated it instead as a 3-variable relation $M(x, y, z)$), then our previous papers [25, 26] have established that significant exceptions to Gödel's Second Incompleteness Theorem can be found for the Semantic Tableaux deductive calculi. Thus, the combination of these prior exceptions to the Second

Incompleteness Theorem, together with our new generalization of it, will establish that it is the very act of changing Multiplication from a 3-variable relation to a total function which is the exact juncture point where the Semantic-Tableaux version of Gödel's Second Incompleteness Theorem becomes valid. (The exact nature of this tight match between complementing positive and negative results will be explored in great detail by Sections 5 and 6.)

Another consequence of our formalism is that it will imply that the semantic-tableaux version of the Second Incompleteness Theorem generalizes fully for the axiom system IS_0 , as well as for all its natural extensions. (This answers an open question raised twenty years ago by Paris and Wilkie [15].)

2 NOTATION AND INTUITION

The results of this paper will hold for all the main conventional definitions of a Semantic Tableaux. For simplicity, our particular definition of this deductive calculi will be very similar to the precise definitions appearing in Smullyan's and Fitting's textbooks [7, 20]. Thus a **Semantic Tableaux Proof** of a theorem Φ will consist of a tree whose root is the sentence $\neg\Phi$ and whose all other nodes are either axioms of α or deductions from higher nodes of the tree. Let the notation " $\mathcal{A} \mapsto \mathcal{B}$ " indicate that $\mathcal{B}$ is a valid deduction when $\mathcal{A}$ is an ancestor of $\mathcal{B}$ in the proof-tree T . In this notation, the deduction rules allowed under our definition of Semantic Tableaux are:

1. $\Upsilon \wedge \Gamma \mapsto \Upsilon$ and $\Upsilon \wedge \Gamma \mapsto \Gamma$.
2. $\neg\neg\Upsilon \mapsto \Upsilon$. Other valid Tableaux Elimination-Rules for the " $\neg$ " symbol includes the rules:
 $\neg(\Upsilon \vee \Gamma) \mapsto \neg\Upsilon \wedge \neg\Gamma$, $\neg(\Upsilon \Rightarrow \Gamma) \mapsto \Upsilon \wedge \neg\Gamma$, $\neg(\Upsilon \wedge \Gamma) \mapsto \neg\Upsilon \vee \neg\Gamma$,
 $\neg\exists v \Upsilon(v) \mapsto \forall v \neg\Upsilon(v)$ and $\neg\forall v \Upsilon(v) \mapsto \exists v \neg\Upsilon(v)$.
3. A pair of sibling nodes Υ and Γ is allowed in a candidate tree when their ancestor is $\Upsilon \vee \Gamma$.
4. Similarly. a pair of sibling nodes $\neg\Upsilon$ and Γ is allowed in a candidate tree when their ancestor is $\Upsilon \Rightarrow \Gamma$.
5. $\exists v \Upsilon(v) \mapsto \Upsilon(u)$ where u denotes a *newly introduced* psuedo-constant symbol.
6. $\forall v_1 \forall v_2 \dots \forall v_k \Upsilon(v_1, v_2 \dots v_k) \mapsto \Upsilon(t_1, t_2 \dots t_k)$ where each t_i denotes a term free in Υ .

Define a particular leaf-to-root branch in a candidate tree T to be **Closed** iff it contains both some sentence Υ and its negation $\neg\Upsilon$. Then following Smullyan's convention [7, 20], a **Semantic Tableaux** proof of Φ shall be defined to be a tree whose root stores the sentence $\neg\Phi$ and whose every root-to-leaf branch is closed.

For simplicity, our base language will be the same as the language of Robinson's Arithmetic Q. Thus only Addition and Multiplication function symbols will appear in its terms t , and the existential and universal quantifiers in " $\exists x \leq t \phi(x)$ " and " $\forall x \leq t \phi(x)$ " will be called *bounded quantifiers*. A formula

will be called Δ_0 if all its quantifiers are bounded. Similarly if $\Psi(x_1, x_2, \dots, x_k)$ is Δ_0 , then we will use the term Π_1 for describing the prenex-normalized sentence $\forall x_1, \forall x_2 \dots \forall x_k \Psi(x_1, x_2, \dots, x_k)$.

In our discussion, $\text{SemPrf}_\alpha(x, y)$ will denote a Δ_0 formula indicating that y is a semantic tableaux proof of the theorem x from the axiom system α . The first paragraph of this section gave our definition of “Semantic Tableaux”, but with one minor caveat, our version of the Second Incompleteness Theorem will actually generalize to all the other major definitions of semantic tableaux as well. The caveat is that each different definition of semantic tableaux deduction will be associated with a slightly different Π_1 sentence V such that $Q + V$ is a threshold for the Second Incompleteness Theorem. (This is essentially because the Π_1 sentence V , constructed in the next section, will contain a Δ_0 subformula “SemPrf” that is dependent on the particular definition of Semantic Tableaux proof used.)

Also, our version of the Second Incompleteness Theorem will generalize for all the other common cut-free methods of deduction (such as for example Herbrand Deduction, Resolution and the Cut-Free Sequent Calculus) except for the same caveat that for each corresponding deduction method d , a slightly different Π_1 sentence V_d will formally replace V .

Coding Conventions. Given a sequence of integers $i_1, i_2, i_3, \dots, i_m$, we will say a Δ_0 formula $\Phi(g, j, x)$ makes g an encoding of this sequence iff $\Phi(g, j, x)$ is satisfied only when x represents the j -th element in the sequence $i_1, i_2, i_3, \dots, i_m$. We will say g is a **Linear Compressed Encoding** of this sequence if $\text{Log}(g)$ has a magnitude proportional to the size of $\sum_{j=1}^m \text{Log}(i_j + 2)$. Buss, Hájek, Paris, Pudlák and Wilkie [6, 9, 24] have all given examples of such Δ_0 encodings, and we will therefore not also do so here. The Linear Compressed Encodings are considered to be the most efficient possible method to do a formal Gödel encoding of a sentence or proof, and we will therefore study it.

Let the symbol $\perp$ denote the Gödel number of the sentence $0 = 1$. The *weakest possible definition* of α 's Semantic Tableaux Consistency is the assertion:

$$\forall p \neg \text{SemPrf}_\alpha(\perp, p) \quad (3)$$

If $\text{Def}(\alpha)$ denotes the above definition of α 's tableaux consistency then there are also other available definitions, denoted as say $\text{Def}^*(\alpha)$, such that the two definitions are equivalent in adequately strong fragments of arithmetic, *but not in sufficiently weak fragments* (see for instance [26, 27, 28]). It is preferable to employ the *weakest available definition* when generalizing the Second Incompleteness Theorems. This is why our present article uses Equation (3)'s definition.

Although our main goal will certainly be to prove the Second Incompleteness Theorems using the definition of consistency given in Equation (3), our mathematical machinery for establishing this result will run much more smoothly if we use *a different definition of a proof* during the intermediate steps of the analysis. Therefore, the remainder of this paper will also use the following notation:

Definition 2.1. Let $\text{Log}(x)$ denote Base-2 Logarithm, with *downwards rounding* to the lowest integer. (We shall assume $\text{Log}(0) = 0$.) Let $\text{Log}(x, k)$ denote $\text{Log}(\text{Log}(\text{Log} \dots (\text{Log}(x))))$ – where there are k iterations of logarithm. For any fixed constant $K > 1$, the symbol $\text{SemPrf}_\alpha^K(x, y, z)$ will denote a Δ_0 formula indicating that $\text{SemPrf}_\alpha(x, y)$ is valid **AND** that $y < \text{Log}(z, K)$.

Lemma 3.1 will explain how to encode the graphs of $\text{Log}(x, k)$ and hence $\text{SemPrf}_\alpha^K(x, y, z)$ as Δ_0 formulae. This lemma is postponed until the next section because the intuition behind our general framework is better explained with Lemma 3.1's proof postponed.

The role that Definition 2.1 will play in the main proofs in this paper theorems will be quite odd. Our main analysis will employ a detour, where Definition 2.1's extra layer of notation (above) is helpful because our unusually weak axiom systems α will not assume that any function growing faster than Multiplication is a total function. To compensate for this weakness of α , we will often rely upon “ $\text{SemPrf}_\alpha^K(x, y, z)$ ” instead of “ $\text{SemPrf}_\alpha(x, y)$ ” to establish the local existence of numbers sufficiently larger than y .

We will undertake a special effort in this paper to assure that our final generalizations of the Second Incompleteness theorem will concern exclusively the more desirable *conventional* definition of tableaux-consistency given in Equation (3) (rather than some artificial analog that has Definition 2.1's $\text{SemPrf}_\alpha^K(x, y, z)$ formula replace $\text{SemPrf}_\alpha(x, y)$). Our basic strategy will be to employ the artificial formula $\text{SemPrf}_\alpha^K(x, y, z)$ only as a handy device to shorten some intermediate parts of our analysis.

More Details: Let $D(\alpha)$ denote Gödel's famous diagonalization sentence (formally defined below):

* There is no Semantic Tableaux proof of **this sentence**.

Also, let $D^K(\alpha)$ denote the obvious $\text{SemPrf}_\alpha^K(x, y, z)$ modification of this sentence (defined below):

** In a context where one employs the slightly modified “ $\text{SemPrf}_\alpha^K(x, y, z)$ ” proof-notation, there exists no code (y, z) that “proves” **this sentence**.

It is very easy to formally encode $D^K(\alpha)$ as a Π_1 sentence following the example of the prior literature on diagonalization. Thus, let $\text{Subst}(g, h)$ denote Gödel's classic Δ_0 substitution relation, defined below:

$\text{Subst}(g, h) =$ The integer g is an encoding of a formula, and h encodes a sentence identical to g , except that all free variables in g are replaced with a constant, whose value equals g .

Then $D^K(\alpha)$ can be defined as being the Π_1 sentence $\Gamma(\bar{n})$, where $\Gamma(g)$ denotes the formula (4) and $\bar{n}$ denotes $\Gamma(g)$'s Gödel number.

$$\forall h \forall y \forall z \quad \{ \text{Subst}(g, h) \Rightarrow \neg \text{SemPrf}_\alpha^K(h, y, z) \} \quad (4)$$

In essence, our version of a proof of the Second Incompleteness Theorem will be similar to the classic diagonalization proofs, except that many intermediate steps will use $D^K(\alpha)$ instead of $D(\alpha)$. One would

first naturally suspect that there must be some serious penalty for letting the unconventional diagonalizing sentence $D^K(\alpha)$ replace $D(\alpha)$. However, it will turn out that there is no serious disadvantage for this change because it is only our intermediate steps (rather than our final theorems) that will be affected by this change.

Theorem 2.2: (*A trivial corollary to Gentzen's Cut Elimination Theorem that we will often employ:*)

Let $\alpha \vdash_S \Lambda$ denote that there is a semantic tableaux proof of the theorem Λ from the axiom system α . Then the combination of $\alpha \vdash_S \Lambda$, $\alpha \vdash_S \Theta$, and $\alpha \vdash_S \Lambda \wedge \Theta \Rightarrow \Xi$ implies $\alpha \vdash_S \Xi$.

Proof Sketch: Let p, q and r denote the proofs of Λ , Θ and $\Lambda \wedge \Theta \Rightarrow \Xi$. If S had designated the Hilbert proof-method, then one could construct the proof t of Ξ by essentially concatenating together the three initial proofs p, q and r . For cut free proof methods, such as Semantic Tableaux, the construction of t is more complicated (and t 's length can certainly be super-exponentially longer than the combined lengths of p, q and r). However, t 's actual existence is assured by Gentzen's Cut Elimination Theorem [8, 23]. $\square$

Theorem 2.3: (*An easy corollary to Gödel's Incompleteness Theorem:*) Suppose α is a finite extension of Q that (for some constant K) proves the three theorems below. Then α is inconsistent.

- A) $\forall p \neg \text{SemPrf}_\alpha(\perp, p)$
- B) $\{ \exists y \exists z \text{SemPrf}_\alpha^K(\lceil D^K(\alpha) \rceil, y, z) \} \Rightarrow \exists x \text{SemPrf}_\alpha(\perp, x)$
- C) $\forall g \forall h \forall h^* \{ \text{Subst}(g, h) \wedge \text{Subst}(g, h^*) \} \Rightarrow h = h^*$

Proof of Theorem 2.3 The justification of Theorem 2.3 is virtually identical to one of Gödel's classic diagonalization proofs. Let D^* denote the “diagonalizing” sentence defined below:

$$D^* =_{\text{df}} \{ \forall y \forall z \neg \text{SemPrf}_\alpha^K(\lceil D^K(\alpha) \rceil, y, z) \} \quad (5)$$

We will apply Theorem 2.2 several times to help shorten the proof of Theorem 2.3. First, let us apply Theorem 2.2 with Λ and Θ denoting the sentences from (A) and (B) in Theorem 2.3's hypothesis and with Ξ representing the sentence D^* (from Equation (5)). For these three particular values for Λ , Θ and Ξ , it is immediate that $\alpha \vdash_S \Lambda \wedge \Theta \Rightarrow \Xi$. Hence, Theorem 2.2 implies

$$\alpha \vdash_S D^* \quad (6)$$

For any fixed K , $D^K(\alpha)$ and D^* are certainly equivalent sentences in sufficiently strong models of Arithmetic. However, we need more than this fact to duplicate Gödel's diagonalization proof in the present setting. We need that *the weak axiom system α is yet strong enough* to also recognize this equivalence!

To establish this fact, we begin by recalling that $\bar{n}$ denotes (4)'s Gödel number and that $D^K(\alpha)$'s definition implies $\text{Subst}(\bar{n}, \lceil D^K(\alpha) \rceil)$ is true. Since $\text{Subst}(\bar{n}, \lceil D^K(\alpha) \rceil)$ is a valid Δ_0 sentence and

since Robinson's Arithmetic Q can prove all valid Δ_0 sentences, it follows that Q can prove this sentence. Thus since Theorem 2.3's hypothesis indicates that α is an extension of Q, we get:

$$\alpha \vdash_S \text{Subst}(\bar{n}, [D^K(\alpha)]) \quad (7)$$

We will now use Equation (7) to infer the validity of (8) below. In particular, we do so by applying Theorem 2.2 with Λ representing the sentence $\text{Subst}(\bar{n}, [D^K(\alpha)])$, with Θ representing the sentence (C) in Theorem 2.3's hypothesis, and with Ξ being the identity $D^K(\alpha) \equiv D^*$. For these three values for Λ , Θ and Ξ , it is easy to infer that $\alpha \vdash_S \Lambda \wedge \Theta \Rightarrow \Xi$ (because $\Lambda \wedge \Theta$ enables α to immediately deduce that the only value for h satisfying the left side of the implication clause from Equation (4) is the quantity $[D^K(\alpha)]$). Hence, Theorem 2.2 implies:

$$\alpha \vdash_S D^K(\alpha) \equiv D^* \quad (8)$$

Our third and last application of Theorem 2.2 is entirely trivial. We set Λ and Θ equal to the sentences D^* and $D^K(\alpha) \equiv D^*$. Then from the combination of Equations (6) and (8), we trivially infer that

$$\alpha \vdash_S D^K(\alpha) \quad (9)$$

We will now finish our proof of Theorem 2.3 by following the example of Gödel's paradigm involving proving a sentence that states roughly "*There is no proof of me*". In particular, let p denote the semantic tableaux proof of the theorem $D^K(\alpha)$ (whose existence is assured by Equation (9)). Choose a second integer q satisfying $\text{Log}(q, K) > p$. Let r denote the Gödel number of the sentence $D^K(\alpha)$. Let us also recall that if $\bar{n}$ denotes Equation (4)'s Gödel number, then $D^K(\alpha)$ is formally the sentence:

$$\forall h \forall y \forall z \quad \{ \text{Subst}(\bar{n}, h) \Rightarrow \neg \text{SemPrf}_\alpha^K(h, y, z) \} \quad (10)$$

We follow Gödel's example by observing that (10) must be false because if we replace its three variables y , z , and h with the three constants p , q , and r then (10)'s formal statement is clearly negated. Moreover, Robinson's Arithmetic Q is known to have the capacity to formally disprove any Π_1 sentence that is invalid in the Standard Model of the Natural Numbers. Hence, since α is an extension of Q, we get

$$\alpha \vdash_S \neg D^K(\alpha) \quad (11)$$

The combination of (9) and (11) shows that α is inconsistent. $\square$

Important Clarification About Theorem 2.3's Main Use and Meaning: The superscript K in Theorem 2.3 appeared *only on* the left side of Item B's implication clause. (It was absent from the right side of B's clause and from Item A.) The omission and presence of the superscript K will explain a key part of our planned strategy for using the $D^K(\alpha)$ diagonalization sentence. During the intermediate steps of the next section's analysis, the unconventional $\text{SemPrf}_\alpha^K(x, y, z)$ formalization of a Semantic Tableaux proof will surely be important (since it appears on the left side of Item B's implication clause). On the other hand, because the superscript K is absent elsewhere from Theorem 2.3's formal statement, our final main generalizations of the Second Incompleteness Theorem will have this awkward superscript fully removed. Hence, the basic reason that our final theorems shall pertain to Equation (3)'s fully conventional definition of Semantic Tableaux consistency will be due to Theorem 2.3.

3 THE TWO MAIN THEOREMS

This section will introduce the simplest form of our new version of the Second Incompleteness Theorem. We will use a notation convention where $x - y = 0$ when $x < y$ (so that Subtraction can be viewed as a total function). Also, the function $\text{Log}(x, u)$ was defined by Definition 2.1. Our discussion will begin with several preliminary lemmas.

Lemma 3.1. There exists two Δ_0 formulae $S(x, y, z)$ and $P(x, u, z)$ such that:

1. The graphs of $S(x, y, z)$ and $P(x, u, z)$ represent the set of ordered triples satisfying respectively the Subtraction and Logarithm functions
2. A Π_1 sentence, **henceforth denoted as V_1** , will indicate that the two Δ_0 formulae $S(x, y, z)$ and $P(x, k, z)$ represent total functions assigning Subtraction and Logarithm their usual properties.

Justification of Item 1: Item 1's claims about Subtraction and $S(x, y, z)$ are obviously true. Item 1's further claims about $\text{Log}(x, u)$ and $P(x, u, z)$ are formally proven in the Appendix. (The Appendix's proof is an extension of Benett demonstration [4] that $\text{Log}(x, 1)$ has a Δ_0 encoding. It uses the theory of "LinH functions", explored by Hájek, Krajíček, Pudlák and Wrathall [9, 10, 29], to provide a general methodology for generalizing Benett's result for arbitrary "LinH" functions. The Appendix's proof is an extremely direct consequence of this prior literature.)

Justification of Item 2: The further claim of Item 2 is a mostly trivial consequence of Item 1, but for the sake of completeness Item 2 is proven in the Appendix's concluding paragraph.

Lemma 3.2. Using the predicates $S(x, y, z)$ and $P(x, u, z)$ (from Lemma 3.1), one can easily encode five Π_1 sentences, henceforth denoted as A_1, A_2, A_3, A_4 and A_5 , that indicate that the functions Subtraction and $\text{Log}(x, k)$ have the following five well-known properties:

1. $\forall x \forall y \forall z \quad xy \leq z \Rightarrow \text{Log}(z, 1) \geq \text{Log}(x, 1) + \text{Log}(y, 1)$
2. $\forall x \forall y \quad x^2 \leq y \Rightarrow \text{Log}(y, 2) \geq \text{Log}(x, 2) + 1$
3. $\forall x \forall y \quad \text{Log}(x, y + 1) = \text{Log}(\text{Log}(x, y), 1)$
4. $\forall x \forall y \quad y \leq x \Rightarrow [y = x \vee y \leq x - 1]$
5. $\forall x \forall y \forall z \quad \{z \geq y \Rightarrow z - y = x \leftrightarrow x + y = z\} \wedge \{z \leq y \Rightarrow z - y = 0\}$

Proof: A trivial consequence of Lemma 3.1.

Notation Convention: Let $\text{FinAx}(\alpha)$ denote a Δ_0 formula, which will return the Boolean value of TRUE when the integer α is a Gödel number that represents a finite-length list of logical sentences. We can use this formula as an intermediate step to rewrite the formula " $\text{SemPrf}_\alpha(x, y)$ " so that it can be now treated as a Δ_0 formula with three input variables x, y, α (rather than just x and y). Thus in the next two sections, $\text{SemPrf}_\alpha(x, y)$ will denote a Δ_0 formula that specifies $\text{FinAx}(\alpha)$ is satisfied and that y

represents a semantic proof of the theorem x from the axiom system α . Also, “ $\text{SemPrf}_\alpha^k(x, y, z)$ ” will denote the analogous Δ_0 formula free in the five variables of x, y, z, α and k .

Lemma 3.3. There exists a Δ_0 formula, henceforth denoted as $\text{Map}(\alpha, k, d)$, such that (α, k, d) satisfies this formula if and only if d equals the Gödel number of Section 2’s diagonalization sentence $D^k(\alpha)$. This implies that $\text{Paradox}(y, z, \alpha, k)$, defined below, also has a Δ_0 encoding.

$$\text{Paradox}(y, z, \alpha, k) \quad =_{\text{df}} \quad \exists d < z \quad \{ \text{Map}(\alpha, k, d) \wedge \text{SemPrf}_\alpha^k(d, y, z) \} \quad (12)$$

Justification. The formal proof of Lemma 3.3 is analogous to the Appendix’s proof of Lemma 3.1 in that the relevant Δ_0 formula is constructed by first using a LinH procedure to represent $\text{Map}(\alpha, k, d)$ and then using the fact that every LinH procedure can be transformed into an analogous Δ_0 formula. We will omit giving the further details because the proof is roughly analogous to Lemma 3.1’s proof. (Indeed, its underlying structure is very similar to some proofs from Section 4 of our earlier paper [28].) $\square$

We are now ready to define the “V” used by our axiom system $\text{Q}+\text{V}$. It can be thought of as either one long Π_1 sentence, representing a conjunction of five Π_1 clauses $V_1 \wedge V_2 \wedge V_3 \wedge V_4 \wedge V_5$, or as a list of five different Π_1 axiom-sentences. Its sentence V_1 was defined by Lemma 3.1. In our discussion, $\text{FinAx5}(\alpha)$ will denote a Δ_0 formula indicating α is a Gödel number which encodes some finite list of axioms that includes all the axioms of $\text{Q}+V$. Also, $\text{FinAx4}(\alpha)$ will have an identical definition as $\text{FinAx5}(\alpha)$, except it will require α ’s list of axioms include only $\text{Q}+V_1+V_2+V_3+V_4$. Below are defined V_2, V_3, V_4 and V_5

$$V_2 \quad =_{\text{df}} \quad A_1 \wedge A_2 \wedge A_3 \wedge A_4 \wedge A_5 \wedge A_6 \text{ where Lemma 3.2 defines the first five } A_i, \text{ and } A_6 \text{ states}$$

$$\text{Addition and Multiplication satisfy the Associative, Commutative and Distributive principles} \quad (13)$$

$$V_3 \quad =_{\text{df}} \quad \{ \forall g \forall h \forall h^* \{ \text{Subst}(g, h) \wedge \text{Subst}(g, h^*) \} \Rightarrow h = h^* \} \quad (14)$$

$$V_4 \quad =_{\text{df}} \quad \{ \forall \alpha \forall k \forall g \forall h \forall y \forall z \quad [\Upsilon(\alpha, k, g, h, y, z) \Rightarrow \exists h^* \leq h \exists y^* \leq y \exists z^* \leq z \quad \Upsilon(\alpha, k, g, h^*, y^*, z^*)] \}$$

$$\text{WHERE } \Upsilon(\alpha, k, g, h, y, z) \quad =_{\text{df}} \quad \{ \text{Subst}(g, h) \wedge \text{SemPrf}_\alpha^k(h, y, z) \} \quad (15)$$

CLARIFICATION: Since V_4 is formally provable from Q , some readers will wonder why it is necessary for V_4 to appear as an axiom? The answer is that a redundant axiom can super-exponentially shorten the length of some cut-free and/or Semantic Tableaux proofs. (This fact will be used by us later to prove Lemma 4.8 by showing how its proof tree x can be adequately compressed.)

$$V_5 \quad =_{\text{df}} \quad \{ \forall y \forall z \forall \alpha \forall k \{ [\text{FinAx4}(\alpha) \wedge k \geq \alpha \wedge \text{Paradox}(y, z, \alpha, k)] \Rightarrow \exists x < z \text{SemPrf}_\alpha(\perp, x) \} \} \quad (16)$$

Below are listed our two main theorems about $Q+V$.

Theorem 3.4: The axiom system $Q+V$ is consistent.

Theorem 3.5: Suppose α is a consistent axiom system that is a finite extension of $Q+V$. Then α cannot prove a theorem asserting its Semantic Tableaux consistency (i.e. it cannot verify $\forall x \neg \text{SemPrf}_\alpha(\perp, x)$).

Comment 3.6: Before presenting the proofs of Theorems 3.4 and 3.5, it is desirable to briefly reflect upon our overall goals and strategies. The chief goal is, of course, to establish the existence of some Π_1 sentence V where $Q + V$ satisfies simultaneously Theorem 3.4's consistency property and Theorem 3.5's Incompleteness property. It turns out that there are many different Π_1 sentences V that possess both these properties. Some of these V will oddly cause Theorem 3.4's half of the proof to be shorter, while others will cause Theorem 3.5's portion of the proof to be shorter. After some experimentation with this subject, we discovered that the paper's overall presentation would be best abbreviated if we chose to work with a particular axiom-sentence V where Theorem 3.5's portion of the proof would become extremely short, but Theorem 3.4's part of the proof would become much longer. The reason we are informing our readers about this trade-off is that some readers might otherwise find Theorem 3.5's proof (below) to be so alarmingly abbreviated to make it initially quite confusing. The best way to soothe such concerns is to remind our readers that it is possible to find some Π_1 sentences V , where $Q + V$'s Incompleteness property has an astonishingly short proof, provided Theorem 3.4's proof of $Q + V$'s consistency property then becomes correspondingly longer.

Proof of Theorem 3.5 Suppose for the sake of contradiction the theorem was false. Then there would exist some fixed integer constant, denoted as say $\bar{\alpha}$, such that $\text{FinAx5}(\bar{\alpha})$ is true and where $\bar{\alpha}$, viewed as an axiom system, is consistent and verifies its own Semantic-Tableaux-Consistency.

Hence, we may begin our contradiction proof by letting $\bar{\alpha}$ represent a consistent axiom system satisfying

$$\bar{\alpha} \vdash \forall x \neg \text{SemPrf}_{\bar{\alpha}}(\perp, x) \quad (17)$$

Our goal is to show how Equation (17) will lead to a contradiction.

Let us now introduce a second constant $\bar{k} = \bar{\alpha} + 1$. We claim that the ordered pair $(\bar{\alpha}, \bar{k})$ will satisfy the hypothesis of Theorem 2.3. To establish this fact, we must show that $\bar{\alpha}$ can prove the three theorems (A), (B) and (C) required by Theorem 2.3's hypothesis. Below is the justification for this claim:

1. Equation (17) shows that $\bar{\alpha}$ can prove the sentence (A) from Theorem 2.3's hypothesis.
2. We will next show that $\bar{\alpha}$ can also prove Theorem 2.3's sentence (B). Let Λ and Θ denote the following two sentences:

$$\text{FinAx4}(\bar{\alpha}) \wedge \bar{k} \geq \bar{\alpha} \wedge \text{Map}(\bar{\alpha}, \bar{k}, [D^{\bar{k}}(\bar{\alpha})])$$

$$\forall y \forall z \{ [\text{FinAx4}(\bar{\alpha}) \wedge \bar{k} \geq \bar{\alpha} \wedge \text{Paradox}(y, z, \bar{\alpha}, \bar{k})] \Rightarrow \exists x < z \text{SemPrf}_{\bar{\alpha}}(\perp, x) \}$$

Both these sentences are provable from $\bar{\alpha}$. In particular, $\bar{\alpha} \vdash \Lambda$ holds because $\bar{\alpha}$ can prove every Δ_0 sentence that is valid in the Standard Model of the Natural Numbers, and $\bar{\alpha} \vdash \Theta$ holds because Θ 's formal statement is identical to $\bar{\alpha}$'s V_5 axiom except that V_5 's universally quantified variables α and k are replaced by the constants $\bar{\alpha}$ and $\bar{k}$. Moreover, if we let Ξ denote the sentence (B) from Theorem 2.3's hypothesis, it is straightforward to obtain $\bar{\alpha} \vdash \Lambda \wedge \Theta \Rightarrow \Xi$ (see footnote¹). Hence from Theorem 2.2, we immediately obtain that $\bar{\alpha}$ can also prove statement (B).

3. Since Theorem 2.3's sentence (C) and $\bar{\alpha}$'s axiom V_3 are the same sentence, $\bar{\alpha}$ can trivially prove (C)

Hence since $\bar{\alpha}$ satisfies Theorem 2.3's three requirements, it must be inconsistent. This observation completes our proof-by-contradiction because its first paragraph had assumed $\bar{\alpha}$'s consistency. $\square$

4 THE CONSISTENCY OF Q+V

The axiom system Q+V and the preceding Theorem 3.5 would obviously both be entirely useless if Q+V was actually inconsistent. Therefore to establish their significance, we must prove Q+V's consistency.

Our proof of Theorem 3.4 will rest on showing that all five of V 's axioms are valid in the Standard Model. The proof will begin with several preliminary lemmas.

Lemma 4.1. The first four axioms of V are valid in the Standard Model of the Natural Numbers, and these axioms can be written as Π_1 sentences.

Proof. Lemmas 3.1 and 3.2 imply the axioms V_1 and V_2 are valid Π_1 sentences in the Standard Model of the Natural Numbers. It is trivial that V_3 is also a valid Π_1 sentence. Finally for any Δ_0 formula $\phi(x, y)$, the sentence “ $\forall d \forall e [\phi(d, e) \Rightarrow \exists f \leq e \phi(d, f)]$ ” is clearly a valid Π_1 sentence. This implies that V_4 is also a valid Π_1 sentence, since its Equation (15) has a form identical to the preceding sentence except that it replaces d with (α, k, g) , e with (h, y, z) and f with (h^*, y^*, z^*) . $\square$

The remainder of this section will finish the proof of Theorem 3.4 by showing that the axiom V_5 satisfies a logical validity property similar to V_1 through V_4 .

Lemma 4.2. If the clause on the left side of the axiom V_5 's implication symbol is true then $y < \text{Log}(z, 2^{1,000})$.

¹We have inserted our proof of $\bar{\alpha} \vdash \Lambda \wedge \Theta \Rightarrow \Xi$ into a footnote because this identity can be proven using quite standard methods. In particular, the formal statement of Ξ was given in Item B of Theorem 2.3, and it is duplicated below:

$$\{ \exists y \exists z \text{ SemPrf}_\alpha^K ([D^K(\alpha)] , y, z) \} \Rightarrow \exists x \text{ SemPrf}_\alpha (\perp , x)$$

Let $\text{LS}(\Xi)$ and $\text{RS}(\Xi)$ denote the subformulae on the left and right sides of the above horn clause. From the definition of $\text{Paradox}(y, z, \alpha, k)$ (given in Equation (12)), it is obvious that the identity $\Lambda \wedge \Theta \wedge \text{LS}(\Xi) \Rightarrow \text{RS}(\Xi)$ holds in all models of Q . Hence so must the identity $\Lambda \wedge \Theta \Rightarrow \Xi$ also hold in all models of Q . Since Semantic Tableaux proofs are known to satisfy the model-completeness property, this implies there must exist a semantic tableaux proof from Q of the theorem $\Lambda \wedge \Theta \Rightarrow \Xi$. Since $\bar{\alpha}$ is an extension of Q , the latter obviously also implies there exists a semantic tableaux proof of $\Lambda \wedge \Theta \Rightarrow \Xi$ from $\bar{\alpha}$.

Proof. The formula on the left side of the axiom V_5 's $\Rightarrow$ symbol is:

$$\text{FinAx4}(\alpha) \wedge k \geq \alpha \wedge \text{Paradox}(y, z, \alpha, k) \quad (18)$$

Equation (18)'s FinAx4 formula indicates that $\alpha \supseteq Q + V_1 + V_2 + V_3 + V_4$. Clearly $k \geq 2^{1,000}$ because (18) indicates that $k \geq \alpha$ and α must require substantially more than 1,000 bits to encode an axiom system that includes the full entirety of $Q + V_1 + V_2 + V_3 + V_4$. Moreover since (18) indicates that $\text{Paradox}(y, z, \alpha, k)$ is satisfied, Equation (12) and Definition 2.1 imply $y < \text{Log}(z, k)$. Hence, the last two inequalities also imply $y < \text{Log}(z, 2^{1,000})$. $\square$

Definition 4.3. The symbol “ $\underbrace{N}$ ”, will denote the **canonical binary representation** of the natural number N . It will be a term of length $O(\text{Log } N)$ that defines the value of N using only the constant symbols for the numbers 0, 1 and 2. In particular, let $b_0, b_1, \dots, b_m$ denote a sequence of bits where $N = \sum_{i=0}^m b_i 2^i$ and $b_m = 1$. Then $\underbrace{N}$ is defined as:

$$(b_0 + 2 \cdot (b_1 + 2 \cdot (b_2 + 2 \cdot (\dots (b_{m-1} + 2 \cdot b_m))))))$$

Lemma 4.4. Let $\phi(x_1, x_2 \dots x_m)$ denote a Δ_0 formula. For any m-tuple $(c_1, c_2 \dots c_m)$ with each $c_i \leq n$, assume that the semantic tableaux proof (or disproof) of $\phi(\underbrace{c_1}, \underbrace{c_2} \dots \underbrace{c_m})$ from the axiom system α requires a tree with no more than s nodes. Let the symbol “ $Q_i(x_i \leq \underbrace{k_i})$ ” be an abbreviation for either a bounded existential or universal quantifier. Let Ψ denote a canonical Δ_0 sentence of the following form:

$$Q_1(x_1 \leq \underbrace{k_1}), Q_2(x_2 \leq \underbrace{k_2}) \dots Q_m(x_m \leq \underbrace{k_m}) \{ \phi(x_1, x_2 \dots x_m) \} \quad (19)$$

Assume α is an extension of $Q + V_1 + V_2$ and $n \geq \text{MAX}(k_1, k_2, \dots, k_m)$. Then it follows that:

1. If the sentence Ψ (formally defined by Equation (19)) is TRUE then there will exist a semantic tableaux proof of it of approximate size $O(s \cdot n^{2m})$.
2. And if the sentence Ψ is FALSE then there will analogously exist a semantic tableaux disproof of it of approximate size $O(s \cdot n^{2m})$.

Proof Sketch. Since all the quantifiers in Equation (19) are bounded quantifiers, it is immediately apparent that a brute-force computerized-search algorithm can span the cross-product space of approximate size $O(n^m)$ and determine whether or not the sentence (19) is true or false. The fundamental point is that if an axiom system α is rich enough to include all the axioms of $Q + V_1 + V_2$, then a corresponding semantic tableaux proof from α can simulate this exhaustive search and produce either a proof or disproof of (19) using a proof-tree of approximately the same length. In particular, a semantic proof of size $O(s \cdot n^{2m})$ will be certainly adequate for simulating the brute-force algorithm. (Actually if one cares to be meticulous, it can be established that smaller lengths are also adequate.) We will not provide a formal proof of Lemma 4.4 here because the remaining details are straightforward, and the preceding proof-sketch should be adequate to explain its main intuition.

Definition 4.5. Given an axiom system α , say q is a **Closed Subtree Rooted in Ψ** iff q has a structure identical to a semantic tableaux proof, *except that its root consists of “ Ψ ”* (rather than “ $\neg \Psi$ ”).

Lemma 4.6. Let $\underbrace{\alpha}$, $\underbrace{k}$ and $\underbrace{g}$ denote the canonical binary representations of any three numbers, and consider the following sentence:

$$\forall h \forall y \forall z \quad \{ \text{Subst}(\underbrace{g}, h) \Rightarrow \neg \text{SemPrf}_{\underbrace{\alpha}}^{\underbrace{k}}(h, y, z) \} \quad (20)$$

Suppose p is a Semantic Tableaux proof from α of the theorem (20). Let Υ denote the formula from Equation (15). Then for any three terms t_1 , t_2 and t_3 , it is possible to map p onto a “closed” tableaux subtree q where:

- a. The root of q is “ $\exists h^* \leq t_1 \exists y^* \leq t_2 \exists z^* \leq t_3 \Upsilon(\underbrace{\alpha}, \underbrace{k}, \underbrace{g}, h^*, y^*, z^*)$ ”
- b. There will be a natural correspondence between the paths in p and the corresponding paths in q . Under this correspondence, the resulting path in q will have a length exceeding the path in p by no more than some constant C whose value is independent of α, k, g, h, y and z . (Different authors use slightly different definitions of a “semantic tableaux deduction”, and these minor variations will produce slightly different values of Lemma 4.6’s constant C . Under a typical definition, such as say that defined in Section 2’s opening paragraph, C can represent a constant roughly $\cong 12$.)

Proof: It is obvious that if (20) is provable then the root of q ’s subtree is false (and refutable). Thus, one would intuitively expect to be able to map the proof p onto an analogous tree-proof q of roughly the same height. The formal nature of this transformation is trivial, and omitted. $\square$

Lemma 4.7. Suppose n, z and $e > 2$ satisfy $\text{Log}(z, e) > n$. Let Ψ denote a theorem which states “ $\exists r \text{Log}(r, \underbrace{e}) > \underbrace{n}$ ” (where “Log” is obviously encoded formally using Lemma 3.1’s predicate symbol P). Then there exists a constant C (whose value is independent of n, z and e) such that a semantic tableaux proof t of the theorem Ψ from the axiom system $Q + V$ can have its proof-length bounded by $O\{ \lceil \text{LogLog}(z) \rceil^C \}$.

Proof Sketch. Smullyan’s definition of semantic tableaux [7, 20] implies a proof of the sentence “ $\exists r \text{Log}(r, \underbrace{e}) > \underbrace{n}$ ” will store this sentence’s negation in its root. Thus, the root will be essentially:

$$\forall r \text{Log}(r, \underbrace{e}) \leq \underbrace{n} \quad (21)$$

The remainder of t ’s proof will consist of two fragments, which we shall denote as x_1 and x_0 . The substring x_1 will be the topmost section of the proof-tree t . It will use the fact that $Q + V$ recognizes multiplication as a total function to create a series of newly-created constant symbols $u_0, u_1, u_2, \dots, u_n$, satisfying $u_0 = 2$, $u_{i+1} = (u_i)^2$ and having its last term u_n satisfy $z < u_n \leq z^2$.

The second portion of the proof tree t will be called x_0 . It will use the last paragraph’s $u_n > z$ inequality to formally contradict the root’s sentence (stated in (21)). It is essentially trivial to use the six clauses of the axiom V_2 to construct a formal semantic tableaux proof of this contradiction.

It is also easy to see that the proof substring x_1 will contain $O \{ \text{LogLog}(z) \}$ nodes, and x_0 will contain $O \{ [\text{LogLog}(z)]^C \}$ nodes (for some constant C). In particular, the main function of the six clauses of the axiom V_2 is to assure the proof-tree x_0 will have such a sufficiently small size. (We omit a formal proof justifying this last point because it is quite routine.) The main point is that t will satisfy Lemma 4.7 because the combined sizes of its two fragments, x_0 and x_1 , are simply adequately small. $\square$

Comment About the Proofs of Lemmas 4.4, 4.6 and 4.7. Each of these proofs were abbreviated and omitted certain routine details. One reason for this abbreviated style of presentation was that the remaining details were quite straightforward. However, there was also a second motive for keeping this presentation brief. It is that our main objectives are not centered around *the specific* Π_1 sentence V , defined exactly by Equations (13) through (16). Rather, our chief goal is to study *the more general question* about whether one can construct *any such* Π_1 sentence V where $Q+V$ satisfies the crucial combination of Theorem 3.4's Consistency property and Theorem 3.5's Incompleteness property. The last paragraph of this section will explain why it is unnecessary to examine the proofs of Lemmas 4.5 through 4.7 in overly strenuous details to *understand intuitively* the answer to this *more generalized form of our question*.

Lemma 4.8. Suppose that the 4-tuple (y, z, α, k) satisfies the formula on the left side of the axiom V_5 's $\Rightarrow$ symbol. This formula is duplicated below:

$$\text{FinAx4}(\alpha) \wedge k \geq \alpha \wedge \text{Paradox}(y, z, \alpha, k) \quad (22)$$

Then there will exist some constant C whose value is independent of y, z, α, k such that there exists a semantic tableaux proof x from α of the theorem $0=1$, where x 's bit-length $\leq O \{ [\text{LogLog}(z)]^C \}$.

Notation Used in Lemma 4.8's Proof: During our formal construction of Lemma 4.8's proof tree " x ", we will often place arrows or underbrace signs around the symbols α, k, g, h, y and z . Thus each of the symbols $\alpha, \vec{\alpha}$ and $\underbrace{\alpha}$ will often appear in Lemma 4.8's proof. The first type of symbol will denote the usual variables that appear in the axiom V_5 . (For the reader's convenience, this axiom is duplicated below.)

$$V_5 =_{\text{df}} \{ \forall y \forall z \forall \alpha \forall k \{ [\text{FinAx4}(\alpha) \wedge k \geq \alpha \wedge \text{Paradox}(y, z, \alpha, k)] \Rightarrow \exists x < z \text{SemPrf}_\alpha(\perp, x) \} \} \quad (23)$$

Note that the right-hand part of the axiom V_5 asserts the existence of a proof, called x , of the theorem $0=1$. One of the node sentences lying in the proof x will consist of the axiom V_4 . Unfortunately, the axiom V_4 (defined in Equation (15)) contains many of the same variables-names as V_5 (i.e. α, k, y and z). In order to make this notation less ambiguous, we will now put arrow symbols around the variables in Equation (15)'s " V_4 " axiom. Thus, this axiom will now be rewritten as:

$$V_4 =_{\text{df}} \{ \forall \vec{\alpha} \forall \vec{k} \forall \vec{g} \forall \vec{h} \forall \vec{y} \forall \vec{z} [\Upsilon(\vec{\alpha}, \vec{k}, \vec{g}, \vec{h}, \vec{y}, \vec{z}) \Rightarrow \exists h^* \leq \vec{h} \exists y^* \leq \vec{y} \exists z^* \leq \vec{z} \Upsilon(\vec{\alpha}, \vec{k}, \vec{g}, h^*, y^*, z^*)] \} \quad (24)$$

Later in the same Semantic-Tableaux proof x , we will replace the six universally quantified variables, $\vec{\alpha}, \vec{k}, \vec{g}, \vec{h}, \vec{y}$ and $\vec{z}$, from the V_4 axiom with six formal terms. These terms will be written using Definition 4.3's canonical binary formalism, along with its *underbrace notation*.

Proof of Lemma 4.8. Although the statements of Lemmas 4.7 and 4.8 are very different, it will turn out that Lemma 4.8's proof tree x will have some very similar components to Lemma 4.7's tree t . In particular, x will be divided into five fragments, denoted as x_0, x_1, x_2, x_3, x_4 . The first two of these five components will be identical to the x_0 and x_1 fragments of Lemma 4.7's proof tree t .

Some notation will clarify the differences between these two proof trees. If x_i denotes any one of x_0, x_1, x_2, x_3, x_4 , let us use the following terminology:

1. The fragment x_i , viewed as an integer encoding a string of bits, will be said to be **z-tiny** iff $\text{Log}(x_i) \leq O(\text{LogLogLog}(z))$ (there are *three* iterations of "Log" attached here to z).
2. The bit-string x_i will be said to be **z-adequately small** iff $\text{Log}(x_i) \leq O\{[\text{LogLog}(z)]^C\}$.

Our proof-tree x will be essentially the concatenation of two "z-adequately small" parts, x_0 and x_1 , with three "z-tiny" subtrees, x_2, x_3 and x_4 . As we already noted, x_0 and x_1 will be identical to their counterparts from Lemma 4.7's tree t . (Thus, it will turn out that the sole differences between the proof-trees, t and x , will be three very minuscule-sized "z-tiny" fragments.)

We will now formally describe the proof-tree x . Since x is a proof of the theorem $0=1$, the root of its proof-tree will obviously be " $0 \neq 1$ ". Immediately below this root will be the fragment x_1 from Lemma 4.7's proof. It will thus consist of an iterated series of newly-created constant symbols $u_0, u_1, u_2, \dots, u_n$, satisfying $u_0 = 2$, $u_{i+1} = (u_i)^2$ and having its last term u_n satisfy

$$z < u_n \leq z^2 \quad (25)$$

At the bottom of the proof fragment x_1 will appear two further sentences. The first will be Equation (24)'s axiom V_4 . (It is permissible to include this axiom in the proof x because Equation (22)'s $\text{FinAx4}(\alpha)$ clause indicates that the axiom system α includes this axiom.) The last sentence at the bottom of the segment x_1 is formally defined by Equation (26) at the end of this paragraph. This last sentence will be identical to the axiom V_4 , except that V_4 's six universally quantified variables will now be replaced by six terms, denoted as $\underbrace{\alpha}, \underbrace{k}, \underbrace{g}, \underbrace{h}, \underbrace{y}$ and u_n . The following rules will define these terms:

1. Let us recall that Lemma 4.8's hypothesis indicated that (y, z, α, k) satisfied Equation (22). The values of $\underbrace{\alpha}$, $\underbrace{k}$ and $\underbrace{y}$ will represent the corresponding quantities in the tuple (y, z, α, k) . Formally, these three terms will be encoded using the "Canonical Binary" form of Definition 4.3. Also, the tuple (y, z, α, k) will define u_n 's value because u_n satisfies Equation (25).
2. Let us recall that Equation (22)'s $\text{Paradox}(y, z, \alpha, k)$ formula indicates that y is a proof of the theorem $D^k(\alpha)$. Let h denote $D^k(\alpha)$'s Gödel number. We saw previously in Equation (4) of Section 2 how h was constructed. (It basically was constructed using methods similar to Gödel's construction of a sentence that states "*There is no proof of me*". In particular, h was constructed by taking a "masking formula" g , that was free in one variable, and letting h denote the unique integer that satisfies the Gödel-like formula of $\text{Subst}(g, h)$.) Our formal definition of the terms $\underbrace{g}$ and $\underbrace{h}$ is that they will be canonical binary terms that represent the particular integer values associated with these two numbers g and h .

Thus using the definitions above for $\underbrace{\alpha}$, $\underbrace{k}$, $\underbrace{g}$, $\underbrace{h}$, $\underbrace{y}$ and u_n , the node immediately below V_4 in our proof tree will be identical to V_4 , except that V_4 's six universally quantified variables of $\vec{\alpha}$, $\vec{k}$, $\vec{g}$, $\vec{h}$, $\vec{y}$ and $\vec{z}$ will be replaced by these corresponding terms. This sentence is shown below.

$$\Upsilon(\underbrace{\alpha}, \underbrace{k}, \underbrace{g}, \underbrace{h}, \underbrace{y}, u_n) \Rightarrow \exists h^* \leq \underbrace{h} \quad \exists y^* \leq \underbrace{y} \quad \exists z^* \leq u_n \quad \Upsilon(\underbrace{\alpha}, \underbrace{k}, \underbrace{g}, h^*, y^*, z^*) \quad \} \quad (26)$$

First Branch-Split in the Proof x : Immediately below the node storing the sentence (26) will occur the first branch-split in x 's semantic tableaux proof. This split will be generated by the $\Rightarrow$ Elimination Rule. The two children of (26)'s sentence are therefore given by (27) and (28):

$$\exists h^* \leq \underbrace{h} \quad \exists y^* \leq \underbrace{y} \quad \exists z^* \leq u_n \quad \Upsilon(\underbrace{\alpha}, \underbrace{k}, \underbrace{g}, h^*, y^*, z^*) \quad \} \quad (27)$$

$$\neg \Upsilon(\underbrace{\alpha}, \underbrace{k}, \underbrace{g}, \underbrace{h}, \underbrace{y}, u_n) \quad (28)$$

Construction of the Substring x_2 and the Proof of its z-Tiny Size: The substring x_2 will represent a closed subtree of z-tiny size that is rooted in (27)'s sentence. The reason it is possible to build x_2 is that the hypothesis of Lemma 4.8 indicated that $\text{Paradox}(y, z, \alpha, k)$ was satisfied. From Lemma 4.6A, this implies x_2 exists. Moreover, its size must be z-tiny (because Lemma 4.6B implies $\text{Log}(x_2)$ has the same magnitude as $\text{Log}(y)$ and Lemma 4.2 indicated y was actually much smaller than z-tiny).

The Remaining Fragments of the Proof-Tree x and Their Small sizes: To complete Lemma 4.8's proof, we must show how the subtree descending from (28) is also closed and adequately small. This sentence is the statement “ $\neg \Upsilon(\underbrace{\alpha}, \underbrace{k}, \underbrace{g}, \underbrace{h}, \underbrace{y}, u_n)$ ” where Υ was defined by (15). After applying the Semantic-Tableaux $\neg$ Elimination Rule to the preceding quoted sentence, our remaining part of the proof-tree is rooted in the sentence

$$\neg \text{Subst}(\underbrace{g}, \underbrace{h}) \quad \vee \quad \neg \text{SemPrf}_{\underbrace{\alpha}}^k(\underbrace{h}, \underbrace{y}, u_n) \quad (29)$$

Applying the $\vee$ -Elimination Rule to (29), we get a branch-split generating the sibling nodes:

$$\neg \text{Subst}(\underbrace{g}, \underbrace{h}) \quad (30)$$

$$\neg \text{SemPrf}_{\underbrace{\alpha}}^k(\underbrace{h}, \underbrace{y}, u_n) \quad (31)$$

Moreover from Definition 2.1, it is apparent we can make another branch split below (31). It will generate the following pair of sibling nodes:

$$\neg \text{SemPrf}_{\underbrace{\alpha}}(\underbrace{h}, \underbrace{y}) \quad (32)$$

$$\neg \underbrace{y} < \text{Log}(u_n, \underbrace{k}) \quad \text{where Lemma 3.1's predicate } P \text{ encodes "Log"} \quad (33)$$

Thus to show that the subtree descending from (28) is closed and also sufficiently small, we must analyze the three subtrees that descend from the corresponding sentences (30), (32) and (33). These three subtrees will be called x_3 , x_4 and x_0 , and their size analysis is given below:

1. The quantities g and h from (30)'s formula “ $\text{Subst}(\underbrace{g}, \underbrace{h})$ ” must be integers less than y (simply because y is a proof of the theorem whose Gödel number = h). Since, Subst is a Δ_0 formula, it then follows from Lemma 4.4 that the subtree x_3 descending from (30) has a size approximately equal to y 's magnitude. The final point is that Lemma 4.2 indicated that y was actually much smaller than z -tiny. Hence, the subtree x_3 must certainly have a z -tiny magnitude (since its length is governed by y 's size).
2. The proof that the subtree x_4 descending from (32) is z -tiny is almost identical to Item 1's analysis of x_3 essentially because $\text{SemPrf}_{\alpha}(\underbrace{h}, \underbrace{y})$ is also a Δ_0 formula. Thus once again, Lemma 4.2 allows us to assume that y (and therefore again also h) are sharply smaller than z -tiny. Also α must certainly be z -tiny or smaller (because the combination of Equation (22) and Definition 2.1 imply that $\text{Log}(z, k) > y$, $k \geq \alpha$ and hence $\text{Log}(z, \alpha) > y$). Hence since all three of $\text{SemPrf}_{\alpha}(\underbrace{h}, \underbrace{y})$'s terms are sufficiently small, we can again apply Lemma 4.4 to conclude that the subtree x_4 is z -tiny.
3. The proof that the subtree x_0 descending from (33) is z -adequately small is essentially an immediate consequence of Lemma 4.7. In particular, the tree t , constructed in Lemma 4.7's proof, contained a subtree, which was also called x_0 , whose structure is identical to the object that now descends from (33)'s sentence. Thus by our preceding discussion, x_0 is z -adequately small.

The above observations have completed our proof of Lemma 4.8 because they have shown that each of the five subtrees x_0, x_1, x_2, x_3, x_4 are sufficiently small to satisfy Lemma 4.8 $\square$

Lemma 4.9. The axiom V_5 is valid in the Standard Model of the Natural Numbers.

Intuitive Justification of Lemma 4.9. Almost all the details needed to establish Lemma 4.9 have already appeared in Lemma 4.8's proof. Therefore, we will begin by giving an intuitive sketch of the added functionality needed to complete Lemma 4.9's proof.

The key point is that the formal statement of Lemma 4.8 is almost identical to the formal statement of the axiom V_5 . Thus Lemma 4.8's formal statement indicated that any tuple (y, z, α, k) satisfying the formula (22) can be mapped onto an element x , representing a semantic tableaux proof of $0=1$, whose bit-length is bounded by $O\{[\text{LogLog}(z)]^C\}$. The point is that V_5 's formal statement is the same as that of Lemma 4.8 except that V_5 *instead requires that* $x < z$.

It is relatively easy to strengthen Lemma 4.8's proof to establish this alternate condition. This is intuitively because Lemma 4.8's “ $\text{Log}(x) < O\{[\text{LogLog}(z)]^C\}$ ” asymptote is stronger than V_5 's corresponding inequality “ $x < z$ ” for all but a finite number of elements x . Thus, it will turn out to be fairly easy to revise Lemma 4.8's proof so it will also justify V_5 's validity because we will need to only consider a relatively trivial finite number of additional cases to justify its further claim.

More Detailed Justification of Lemma 4.9: The next several paragraphs are intended for those readers who wish to see in greater detail exactly how we can convert the intuitions (above) into a more formal proof. (Some readers, not interested in these details, may choose to skim the passage below.)

Let D be some constant that satisfies the O-notation of Lemma 4.8's " $O\{ [\text{LogLog}(z)]^C \}$ " asymptote. By this, we simply mean that the constant D should be chosen to be large enough so that Lemma 4.8's $O\{ [\text{LogLog}(z)]^C \}$ asymptotic function is strictly less than the quantity $D \cdot [\text{LogLog}(z)]^C$.

Also, let us use the notation convention defined below:

$$f(z) \stackrel{\text{df}}{=} D \cdot [\text{LogLog}(z)]^C \quad (34)$$

Clearly, there will exist some constant M (whose value depends only on C and D), such that every $z > M$ will automatically have $f(z) < \text{Log}(z)$.

Note that if we rewrite Lemma 4.8 using the f -notation (above), it will state that every tuple (y, z, α, k) satisfying the formula (22) can be mapped onto an element x , representing a semantic tableaux proof of $0=1$, whose bit-length is bounded strictly by $f(z)$. Thus from the preceding paragraph, we can trivially conclude that x 's bit-length will be strictly less than $\text{Log}(z)$ whenever $z > M$.

Moreover, this inequality implies that that all tuples (y, z, α, k) with $z > M$ will automatically satisfy the sentence V_5 's requirements (since it shows that V_5 's requirements on x 's size are less stringent than Lemma 4.8's requirement $\text{Log}(x) < f(z)$ whenever $z > M$).

Thus to complete our proof of validity of the axiom V_5 , we need only consider the alternate case where $z \leq M$. It is necessary for us to show that these z also satisfy V_5 's formal requirements.

It is easy to establish this latter fact because the left side of V_5 's implication clause is the formula:

$$\text{FinAx4}(\alpha) \wedge k \geq \alpha \wedge \text{Paradox}(y, z, \alpha, k) \quad (35)$$

Moreover, Lemma 4.2 indicated that Equation (35) automatically implies:

$$\text{Log}(z, 2^{1,000}) > y \quad (36)$$

We omit the full details (because they are quite trivial), but the lower bound N implied by Equation (36) is a much larger lower bound on z 's permissible values than the alternate lower bound M (associated with Equation (34)). In other words, Equation (36) implies the existence of an integer $N > M$ such that no tuple (y, z, α, k) with $z \leq N$ can possibly satisfy the formula (35).

Every tuple (y, z, α, k) with $z \leq M$ will thus automatically also fail to satisfy the formula (35). All such tuples (with $z \leq M$) will thus vacuously satisfy V_5 's implication clause (simply because (35) corresponds to the left side of V_5 's implication clause).

The latter completes our proof of Lemma 4.9 because we have shown a tuple (y, z, α, k) automatically satisfies V_5 in both the cases where $z > M$ and $z \leq M$. $\square$

Finishing the Proof of Theorem 3.4: The combination of Lemmas 4.1 and 4.9 implies Theorem 3.4's validity because they show all of $Q+V$'s axioms are valid in the Standard Model (hence establishing its consistency). $\square$

Recapitulating This Proof: It is now possible to offer a pleasantly short 2-sentence intuitive summary of Theorem 3.4's proof. The heart of the proof rested on showing the existence of a proof-tree

x that was sufficiently small to assure the validity of the axiom V_5 's requirements for x . The intuitive reason why such an x *must exist* is that its bit-length has the same order of magnitude as Lemma 4.7's proof-tree t , and Lemma 4.7 showed the latter was sufficiently small.

A Strengthened Version of Theorem 3.4: Our preceding discussion had technically only proven that V was a valid sentence in the Standard Model of the Natural Numbers (because that assertion was obviously sufficient to corroborate Theorem 3.4's "consistency claim"). It turns out that a more meticulous analysis will show that our proof can be actually carried out entirely within IS_0 , allowing us to establish the stronger result that V is actually a formal theorem of IS_0 .

How to View the Preceding Proofs: Similar to many scientific papers, some aspects of our proofs of Theorems 3.4 and 3.5 were obviously abbreviated because they were deemed to be trivial. In our present article, these abbreviated parts were actually *mostly irrelevant* to our main objectives, *in addition* to being quite routine. This is because the exact form of the five clauses of the axiom V (defined in Section 3) is extremely far from our main interest. Rather our chief goal (as stated in Section 1) was to *establish the existence of any Π_1 sentence V* where $Q + V$ would satisfy the crucial combination of Theorem 3.4's Consistency property and Theorem 3.5's Incompleteness property. For instance, we could have replaced V_5 's clause " $k \geq \alpha$ " with a stronger " $k \geq \alpha^J$ " (for any constant $J \geq 2$), or have made any of many other countless strengthenings of V by adding other logically valid Π_1 clauses to it. The point is that such revisions of V would enable us to construct alternate Π_1 sentences that would satisfy the same Theorem 3.4 and 3.5 properties as our present V . The point is that we have attempted to abbreviate only those aspects of the last two sections that focussed on *the particular* Π_1 sentence V , defined by equations (13) through (16), rather than on *the general question about whether any such Π_1 sentence* could satisfy Theorems 3.4 and 3.5 simultaneously. Thus aside from being mostly trivial, the remaining abbreviated aspects of the proofs of Lemmas 4.4 through 4.9 should cause a reader no particular concern because they possess little relevance for the "more generalized version" of our problem.

5 Axiom Systems with Infinite Cardinality

We will prove two generalizations of Theorems 3.5 in this section. Our first generalization will extend Theorem 3.5's Second Incompleteness Effect to nearly every axiom system with infinite cardinality. We will need one new definition before this new theorem can be introduced.

Definition 5.1. An axiom system α will be said to satisfy the **Conventional Deciphering Property** iff there exists a *finite subset* of its axioms (henceforth denoted as F) capable of recognizing all of α 's axioms. More precisely in a context of Definition 4.3's canonical binary terms " $\underbrace{N}$ ", this means we can find a *finite* $F \subseteq \alpha$ satisfying the invariant below:

$\forall N$ If N is the Gödel number of one of α 's axioms then F is sufficient for proving this fact about " $\underbrace{N}$ ".

All finite and most infinite extensions of Robinson's System Q satisfy Definition 5.1's requirements. In this context, the following generalization of Theorem 3.5 is seemingly significant.

Theorem 5.2. Suppose α is a consistent axiom system that is an extension of Q+V and that satisfies the Conventional Deciphering Property. Then α cannot prove a theorem asserting its Semantic Tableaux consistency (i.e. it cannot verify the formal statement asserting the non-existence of a Semantic Tableaux proof of $0=1$ from itself).

Proof. Let F be some finite subset of the axiom system α that satisfies Definition 5.1. We will prove Theorem 5.2 with a proof-by-contradiction. Thus for the sake of establishing a contradiction, let us temporarily assume that a consistent system α can prove its own tableaux consistency.

Thus, let E denote the particular finite subset of α 's axioms that is used to prove the formal statement of α 's tableaux consistency. (This means that E is sufficient to prove the non-existence of a Semantic Tableaux proof of $0=1$ from α .) Also, define ζ to be the finite axiom system that is the union of the four axiom groups of E , F , Q and V . Then since ζ includes all the axioms of E , it follows that ζ can also prove α 's tableaux consistency.

Moreover since ζ is finite-sized and includes all the axioms of F , ζ will be able to recognize that each of its own axioms are also axioms of α . This observation implies that ζ will certainly know that α 's tableaux consistency will imply ζ 's tableaux consistency.

Hence the last two paragraphs imply that ζ can certainly prove its own tableaux consistency.

The key point is that ζ must have a finite cardinality because it was defined to be the system $Q + V + E + F$. Thus, Theorem 3.5 must be applicable to ζ simply because of ζ 's finite size. Hence, we can conclude that ζ *must be inconsistent* simply because Theorem 3.5 (once it is rewritten into its logically equivalent contrapositive form) indicates that any finite extension of $Q + V$ that proves its own tableaux consistency is automatically inconsistent.

The preceding argument trivially also establishes that α is inconsistent. (This is because α is an extension of ζ , and the preceding paragraph had shown that ζ was inconsistent.)

The latter fact is sufficient to enable our proof-by-contradiction to now reach its desired conclusion. This is because the opening paragraph of Theorem 5.2's proof had assumed, for the sake of establishing a contradiction, that some **consistent** system α could prove its own Tableaux consistency. We had shown this condition to be impossible in the preceding paragraph when we proved our hypothesis implies that α is inconsistent. Hence, Theorem 5.2 must be valid because its negation is inherently contradictory. $\square$

Let us recall that Gödel defined an axiom system α to be ω -consistent iff it is unable to prove any Σ_1 theorem Ψ that is invalid under the Standard Model of the Natural Numbers. Our next theorem will discuss a type of version of the Second Incompleteness Theorem that is valid for all ω -consistent axiom systems that are extensions of Q. One added definition is necessary before we can introduce this new theorem.

Definition 5.3. Let β , ϕ and p denote the respective Gödel numbers of a finite-sized axiom system, a Π_1 sentence and a semantic tableaux proof. For some fixed constant K to be chosen later, consider a Π_1 sentence of the following form:

$\forall\beta \ \forall\phi \ \forall p$ If β is an extension of Robinson's Q and if p represents a semantic tableaux proof of the theorem “ $0 = 1$ ” from the axiom system $\beta + \phi$, then $\exists r < p^K$ where r represents a semantic tableaux proof of the theorem $\neg\phi$ from β .

For each conventional definition of semantic tableaux deduction, it is obviously possible to choose a large enough constant K to make the sentence above logically valid. (Usually $K = 2$ is good enough, and $K = 2$ is certainly suitable for the particular definition of semantic tableaux that appeared in Section 2's opening paragraph.) Henceforth, we will therefore let $K = 2$ and have V_6 denote the resulting Π_1 sentence above.

Theorem 5.4. Let α denote a finite-sized axiom system and V have the same definition as elsewhere in this paper. Let W denote the Π_1 sentence “ $V \wedge V_6$ ”. Unlike Theorems 3.5 and 5.2, we will now assume that α is an extension of Q (rather than of $Q + V$). Under these assumptions, if the axiom system α is ω -consistent then α will be unable to prove the nonexistence of a Semantic Tableaux proof of the invalid Σ_1 sentence “ $\neg W$ ” from itself.

Proof by Contradiction. Let us assume that Theorem 5.4 was false, and thus an ω -consistent axiom system α could prove the nonexistence of a Semantic Tableaux proof of $\neg W$ from itself. Let α^* denote the union of the axiom system α with this added sentence W . The first sentence of this paragraph, combined with Definition 5.3, then trivially implies (see footnote²) that α^* will have a capacity to prove the nonexistence of a Semantic Tableaux proof of $0 = 1$ from itself.

Note that α^* is an extension of $Q + V$, since $\alpha^* = \alpha + W$. Since α^* formally proves its own tableaux consistency, we may consequently apply Theorem 5.2 (rewritten into its logically equivalent contrapositive form) to conclude that α^* must be inconsistent.

The inconsistency of α^* , together with fact that $\alpha^* = \alpha + W$, then implies that α can prove $\neg W$. Moreover since $\neg W$ is certainly a Σ_1 sentence that is invalid under the Standard Model of the Natural Numbers, this observation also implies α (itself) is ω -inconsistent!

The latter observation completes our proof-by-contradiction because it contradicts the first sentence of Theorem 5.4's proof. (In particular, it contradicts the initial assumption that α was ω -consistent and simultaneously had a capacity to prove the non-existence of a proof of $\neg W$ from itself.) $\square$

Remark 5.5. Our formal statement of Theorem 5.4 had technically required that α be a “finite-sized” axiom system. This requirement was added to Theorem 5.4's hypothesis only for the sake of simplifying its proof. The theorem can actually be strengthened so that it also valid for systems with infinite cardinality

²This is simply because if α^* knows about the non-existence of a proof of the theorem $\neg W$ from α , and if it further contains V_6 as an additional axiom, then it can automatically combine these two facts to immediately infer the non-existence of a proof of the theorem “ $0 = 1$ ” from α^* .

that satisfy Definition 5.1's Conventional Encoding Property. We will not formally prove this generalization of Theorem 5.4 because its formal proof rests on the identical method for reducing an infinite-sized axiom to some finite subcomponent of itself, as was used earlier in Theorem 5.2's proof.

Remark 5.6. Let Q^* denote an axiom system slightly weaker than Robinson's Q in that it will treat Multiplication as a 3-way relation $M(x, y, z)$ rather than as a total function. One reason Theorem 5.4 and Remark 5.5 are interesting is that one can construct examples of ω -consistent extensions of Q^* that have a capacity to prove the non-existence of a proof of $\neg W$ from themselves. For instance, the axiom system $IS^\lambda(A)$ from our article [26] had such a capacity (on account of its "tangibility reflection principle"). Thus, a noteworthy distinction between ω -consistent extensions of Q^* and of Q is that *only the former* are capable of recognizing *the non-existence* of a semantic tableaux proof of proof of $\neg W$ from themselves. Moreover, $IS^\lambda(A)$'s example shows that ω -consistent extensions of Q^* are actually capable of *internally recognizing their own* ω -consistency (see footnote³). Thus, ω -consistent extensions of Q^* have properties quite unlike Theorem 5.4's ω -consistent extensions of Q .

Finally, we wish to remind our readers that the particular definition of Semantic Tableaux deduction used in this article was presented in the opening paragraph of Section 2. From a strictly technical point of view, our Theorems 3.4, 3.5, 5.2 and 5.4 pertained only to this particular definition. However except for one minor caveat, it is possible to generalize all these theorems to basically all the other common definitions of Semantic Tableaux deduction, as well as to other methods of cut-free deduction such as Resolution, Herbrand deduction or the Cut-Free Sequent Calculus. The caveat is quite trivial: It is simply that if we use a different method of cut-free deduction, called say d , then our prior Π_1 sentence V will need to be trivially replaced by a slightly adjusted Π_1 sentence, called say V_d , for the analogs of Theorems 3.4, 3.5, 5.2 and 5.4 to be valid for these alternate d .

6 Two Further Generalizations of Main Result

Some additional notation is needed to establish our last two generalizations of Theorem 3.5. Let us recall that " $\alpha \vdash_S \Phi$," denotes that α has a semantic tableaux proof of the theorem Φ . Also, let us agree that the symbol " $\supseteq$ " in the two phrases of " $\alpha \supseteq \beta$ " and " $\alpha \supseteq \beta + \Upsilon$ " shall have its usual conventional definition. Thus, the first phrase will denote that the axiom system α contains all of the axioms of the system β , and the second phrase will indicate that α contains all of the axioms of β plus the added sentence Υ . In this context, let the symbol " $\geq$ " in the phrase " $\alpha \geq \beta + \Upsilon$ " designate that the following identity holds:

$$\alpha \supseteq \beta \quad \wedge \quad \alpha \vdash_S \Upsilon \tag{37}$$

The Incompleteness properties described by our previous Theorems 3.5 and 5.2 were based essentially on the hypothesis that α satisfied the *containment requirement* " $\alpha \supseteq Q + V$ ". An interesting question

³In particular **FOR EACH** Σ_1 sentence ϕ , $IS^\lambda(A)$'s corresponding localized version of the "Tangibility Reflection Principle" (described in reference [26]) will enable it to *internally verify* the formal theorem statement " $\{ \exists p \text{ SemPrf}_{IS^\lambda(A)}^{\lambda}(\lceil \phi \rceil, p) \} \Rightarrow \phi$ " (which asserts the validity of $IS^\lambda(A)$'s ω -consistency localized relative to ϕ).

is whether or not these theorems would remain valid if we instead employed the weaker assumption “ $\alpha \geq Q + V$ ”.

The answer to the preceding open question is not immediately apparent because a proof of $0=1$ from the axiom system α can become super-exponentially longer under cut-free deduction methods, such as Semantic Tableaux, when we simply replace the requirement “ $\alpha \supseteq Q + V$ ” with the weaker assumption “ $\alpha \geq Q + V$ ”. Our proofs of Lemmas 4.8 and 4.9 would thus break down completely because they required the existence of a very tightly bounded difference between the proof-lengths for the sentence $0=1$ and for the diagonalizing sentence $D^K(\alpha)$. (This tight bound would be violated if the super-exponential growth is allowed to occur.)

In the context of this collapse of Lemmas 4.8 and 4.9, it is not immediately apparent whether or not our main results in Theorems 3.5 and 5.2 will remain valid when we replace their “containment” requirement “ $\alpha \supseteq Q + V$ ” with Equation (37)’s the weaker assumption “ $\alpha \geq Q + V$ ”. However, we can prove two partial positive generalizations of Theorems 3.5 and 5.2 in this context. (One of these will state that the semantic-tableaux version of the Second Incompleteness Theorem generalizes for the axiom system IS_0 .) We will need some further notation to describe these theorems.

Definition 6.1. Let us say that α has a **Hyper-Constructive Semantic Proof** of the theorem Υ iff there exists some sentence, denoted as “ Θ_Υ ”, such that

1. α contains the formal axiom “ $\Theta_\Upsilon \Rightarrow \Upsilon$ ”, and
2. the identity $\alpha \vdash_S \Theta_\Upsilon$ is valid.

Moreover, we will write $\alpha \vdash_H \Upsilon$ when there exists some sentence “ Θ_Υ ” allowing these two conditions to hold.

Definition 6.2. The phrase “ $\alpha \succeq \beta + \Upsilon$ ” will have an identical definition to “ $\alpha \geq \beta + \Upsilon$ ” except that it will replace Equation (37)’s condition “ $\alpha \vdash_S \Upsilon$ ” with Definition 6.1’s construct of “ $\alpha \vdash_H \Upsilon$ ”. Thus, the phrase “ $\alpha \succeq \beta + \Upsilon$ ” is defined as indicating the validity of the identity:

$$\alpha \supseteq \beta \quad \wedge \quad \alpha \vdash_H \Upsilon \tag{38}$$

We will call “ $\alpha \succeq \beta + \Upsilon$ ” the **Hyper-Inclusion Condition**.

It turns out that the hyper-inclusion condition “ $\alpha \succeq \beta + \Upsilon$ ” has properties more similar to the conventional inclusion condition “ $\alpha \supseteq \beta + \Upsilon$ ” than to the inequality “ $\alpha \geq \beta + \Upsilon$ ”. This fact will enable us to establish that the semantic-tableaux version of the Second Incompleteness Theorem generalizes for the axiom system IS_0 . We will begin our discussion with one preliminary lemma.

Lemma 6.3. Suppose that the hyper-proof identity “ $\alpha \vdash_H \Upsilon$ ” holds, and let p denote the proof from α of “ Θ_Υ ” (that is implied to exist by this identity). Let Ψ denote any sentence, and q denote a semantic tableaux proof of the theorem Ψ from the system $\alpha + \Upsilon$. Then there exists a semantic tableaux proof r of the theorem Ψ from the axiom system α such that the difference between the lengths of q and r is an amount that is at most proportional to p ’s length.

Proof. It is easy to construct a semantic tableaux proof r satisfying Lemma 6.3. For instance, we can use the following method to construct r :

1. The root of r will be “ $\neg\Psi$ ” (because Ψ is the theorem which r is seeking to prove).
2. The axiom “ $\Theta_\Upsilon \Rightarrow \Upsilon$ ” will appear immediately below the root. (Its existence is assured by Definition 6.1)
3. A tree-split, using the $\Rightarrow$ Elimination Rule, will appear immediately below this axiom. Its two children will thus be: $\neg\Theta_\Upsilon$ and Υ .
4. From Lemma 6.3’s hypothesis, it follows that a proof-subtree, having p ’s length, can be inserted immediately below the sentence $\neg\Theta_\Upsilon$.
5. Since Lemma 6.3’s hypothesis indicates that there exists a proof q from the axiom system $\alpha + \Upsilon$ of the theorem Ψ , it follows that the same structure can be inserted below the node Υ (because the ancestors of this subtree include the two sentences “ Υ ” and “ $\neg\Psi$ ”).

Thus from Items 4 and 5 above, it follows that r ’s proof-length exceeds q ’s proof-length by an amount that is at most proportional to the length of p . $\square$

We will need to introduce some more notation to establish the validity of the semantic-tableaux version of the Second Incompleteness Theorem for $\text{I}\Sigma_0$. Let us say a Δ_0 formula is **x-focussed** iff it is free only in the variable x . Let us call a Π_1 sentence **x-focussed** iff it can be written in the form “ $\forall x \phi(x)$ ”, where $\phi(x)$ is an x-focussed Δ_0 formula.

We will typically use capitol letter symbols to denote some x-focussed Π_1 sentence. If S is an x-focussed Π_1 sentence of the form say “ $\forall x \phi(x)$ ”, then $\text{Ind}(S)$ will denote the resulting sentence:

$$\phi(0) \wedge \forall x \{ \phi(x) \Rightarrow \phi(x+1) \} \quad (39)$$

The axiom system $\text{I}\Sigma_0$ is defined in for instance [9, 24]. It will contain one axiom of the form (40) for each x-focussed Π_1 sentence S .

$$\text{Ind}(S) \Rightarrow S \quad (40)$$

We will now use this notation to justify the validity of the following theorem:

Theorem 6.4. Let α denote any consistent extension of $\text{I}\Sigma_0$ that satisfies Definition 5.1’s Conventional Deciphering Property. Then α cannot prove the non-existence of a Semantic Tableaux proof of $0=1$ from itself.

Proof Sketch. We will not give a formal proof of Theorem 6.4 because its proof is essentially a repetition of the discussion from Sections 3 through 5, except for containing one extra layer of notation. Instead, we will give an intuitive explanation about how our previous discussion can be easily revised to generate a proof of Theorem 6.4.

The first point is that one can strengthen the proofs of Lemmas 4.1 through 4.9 to show that their analysis can be carried out entirely within the system IS_0 . Thus, IS_0 can corroborate the validity of the sentence V used in our prior $Q + V$ formalism.

The above result means that every axiom system α extending IS_0 satisfies the identity $\alpha \geq Q + V$. However to understand its meaning, we must carefully distinguish between the characteristics of the three different inequalities “ $\alpha \geq Q + V$ ”, “ $\alpha \supseteq Q + V$ ” and “ $\alpha \succeq Q + V$ ”.

In particular, the second paragraph of this chapter already noted the first inequality does not preclude the possibility that some theorem Ψ could have a super-exponentially larger proof from the axiom system α than from $\alpha + V$. Moreover, our prior discussion already explained how such a super-exponential growth, *left unchecked*, could cause our prior proofs of the Second Incompleteness Theorem to break down entirely.

Our basic strategy is to find a sentence V^* , whose properties are identical to the prior V , except that we can replace the old assumption $\text{IS}_0 \vdash_S V$ with the stronger statement that $\text{IS}_0 \vdash_H V^*$. The latter “hyper-constructive” notion of a proof is more desirable because Lemma 6.3 indicated that it was a device for preventing the super-exponential growth in proof length. It will turn out that after we preclude the super-exponential growth, the remainder of our analysis will be essentially identical to our previous discussion (essentially because the absence of a super-exponential growth will cause the resulting inequality “ $\alpha \succeq Q + V^*$ ” to have properties very similar to Section 3’s inclusion relation “ $\alpha \supseteq Q + V^*$ ”). We will explain below more details about how we formalize these concepts.

Formal Details: Given any Π_1 sentence S , it is easy to construct an x-focussed Π_1 sentence S^* , such that IS_0 is capable of recognizing that these two sentences are equivalent. For instance if S is the sentence $\forall u_1 \forall u_2 \dots \forall u_n \psi(u_1, u_2, \dots, u_n)$ then S^* could be:

$$\forall x \forall u_1 \leq x \forall u_2 \leq x \dots \forall u_n \leq x \psi(u_1, u_2, \dots, u_n) \quad (41)$$

Henceforth, we will thus let V^* denote such a rewrite of the Π_1 sentence V into an x-focussed Π_1 form.

Since IS_0 can both prove the validity of V and recognize V ’s equivalence to V^* , it must also have a capacity to prove V^* .

Let us recall that if V^* is a sentence of the form “ $\forall x \phi(x)$ ” then $\text{Ind}(V^*)$ is defined to be a sentence of the form “ $\phi(0) \wedge \forall x \{ \phi(x) \Rightarrow \phi(x+1) \}$ ”. Any proof p of the theorem V^* can thus be trivially modified to also prove $\text{Ind}(V^*)$. (This is because the proof p can be trivially modified to also verify the validity of the two added theorems “ $\phi(0)$ ” and “ $\forall x \phi(x+1)$ ”.) Hence from the preceding paragraph we can trivially conclude that IS_0 can also prove the validity of $\text{Ind}(V^*)$.

The latter observation also enables us to conclude that IS_0 has a hyper-constructive proof of the theorem V^* . (This is because the combination of the facts that $\text{IS}_0 \vdash_S \text{Ind}(V^*)$ and that IS_0 has an

inductive axiom of the form “ $\text{Ind}(V^*) \Rightarrow V^*$ ” implies $\text{IS}_0 \vdash_H V^*$.) Hence, we may apply Lemma 6.3 both to the axiom system IS_0 , as well as to any broader system α satisfying $\alpha \supseteq \text{IS}_0$. It will indicate that if p denotes the hyper-constructive proof of V^* from IS_0 then the following invariant must hold:

An arbitrary theorem Ψ will have a semantic-tableaux-proof q from the system $\alpha + V^*$ only when there exists a proof r of the same theorem Ψ from the system α where the difference between the lengths of q and r is at most proportional to p ’s length.

The significance of the preceding invariant is that it indicates that r ’s length is not super-exponentially larger than q ’s length (as could happen if the hyper-proof construct was absent). The second paragraph of this section had indicated that this super-exponential growth was the chief potential blocking point in our effort to generalize the Second Incompleteness Theorem. The intuitive reason that Theorem 6.4’s generalization of the Second Incompleteness Theorem is valid is thus that the undesirable super-exponential growth cannot occur for any $\alpha \supseteq \text{IS}_0$ because of our invariant’s strong restrictions on the size of r .

We have deliberately avoided proving Theorem 6.4 formally in this section because the remainder of its formal proof is a routine generalization of our prior methodologies. In particular, the footnote ⁴ explains a few added details about how to generalize our proof formalisms from Sections 3-5 to justify Theorem 6.4, but it is probably unnecessary for the reader to examine this footnote to appreciate the main intuition behind Theorem 6.4’s proof. $\square$

Definition 6.5. We will need some added notation to define the sentence V^+ that will be employed by our next theorem. Let us recall that $\text{FinAx4}(\alpha)$ was defined (in Section 3) as a formula indicating that α was a finite-sized axiom system that included all the axioms of **Q** **plus the additional** axioms V_1, V_2, V_3 and V_4 . Let $\text{FinHyp4}(\alpha, x)$ have the same definition as $\text{FinAx4}(\alpha)$ except that it will technically not require that V_1, V_2, V_3 and V_4 to be axioms of α . Instead, it will require that x represent a Hyper-Constructive Semantic proof from the axiom system α of the theorem $V_1 \wedge V_2 \wedge V_3 \wedge V_4$.

Theorem 6.6. There exists a Π_1 sentence V^+ , provable from IS_0 , which assures that every axiom system α satisfying Definition 5.1’s Conventional Deciphering Property will automatically satisfy the following generalization of Gödel’s Second Incompleteness property:

If α is a consistent extension of **Q** that satisfies $\alpha \vdash_H V^+$ then α cannot prove the non-existence of a Semantic Tableaux proof of $0=1$ from itself.

⁴The intuitive reason that Theorem 6.4 must be valid is fairly easy to explain. Note that Section 3’s $\text{SemPrf}_\alpha^K(x, y, z)$ formula guarantees that there is a very large difference between the magnitudes of y and z , since it indicates that $y < \text{Log}(z, K)$. In contrast the difference between the lengths of the two proofs, q and r is defined by the invariant appearing in our indented sentence. It amounts to a relatively small additive constant, proportional to the size of p ’s length. Moreover we can employ *any* fixed constant K when we are applying Theorem 2.3’s diagonalization mechanism, as an intermediate step. It thus follows we can develop a generalized form of the methodologies of Sections 3-5 for proving Theorem 6.4 by choosing a K so large as to assure that the differences between the sizes of y and z will dominate the much smaller difference in magnitudes between q and r . (Once this is done, the remainder of Theorem 6.4’s proof is a routine generalization of our prior methodologies.)

We hope the reader will forgive us for deliberately refraining from giving Theorem 6.4 a fully formal proof. The additional details for generalizing the formalisms of Sections 3-5 to prove Theorem 6.4 are extremely routine, and they constitute the kind of details that one wishes to omit in order to keep a presentation reasonably brief.

Proof Sketch. We will not give a formal proof of Theorem 6.6 because its proof is essentially a repetition of the discussion from Sections 3 through 5, except for containing one extra layer of notation. Instead, we will give an intuitive explanation about how our previous discussion can be easily revised to generate a proof of Theorem 6.6.

Our discussion will not include the formal definition of the sentence V^+ because its definition will be essentially identical to Section 3's axiom V , except for some extremely minor and obvious differences in notation. These notational differences arise essentially because we are now viewing V^+ as a theorem with a hyper-constructive proof, rather than as an axiom of α . (As a result of this change, we have to make some corresponding straightforward changes in V^+ 's formally encoded definition, such as for instance having Definition 6.5's FinHyp4 formula replace Section 3's prior FinAx4 formula.)

After one has formally defined the sentence V^+ , most of the remainder of Theorem 6.6's proof is virtually identical to our prior proof-analysis from Sections 3 through 5. In particular, the only difference will be that one has to apply Lemma 6.3's further formalism to take into account the fact that we are now viewing V^+ as a theorem with a hyper-constructive proof, rather than as an axiom of α . In particular, Lemma 6.3 will assure that V^+ will satisfy the following invariant:

An arbitrary theorem Ψ will have a semantic-tableaux-proof q from the system $\alpha + V^+$ only when there exists a proof r of the same theorem Ψ from the system α , where the difference between the lengths of q and r is an amount at most proportional to the length of the smallest hyper-constructive proof of V^+ .

As had happened in Theorem 6.4's proof, the significance of our invariant is that it will preclude a super-exponential difference between the magnitudes of q and r . Also after precluding this super-exponential jump, it is straightforward to revise the prior formalisms from Sections 3 through 5, line by line once again using our invariant's linearly bounded growth effect, to obtain a formal proof of Theorem 6.6's slightly more elaborate claim. $\square$

We have deliberately kept our proofs of Theorems 6.4 and 6.6 extremely brief because they are essentially routine and quite trivial generalizations of our preceding methodologies. The reason for our interest in Theorem 6.6 is that there is an extremely tight match between its positive generalization of Gödel's Second Incompleteness with the negative results, about exceptions to it mentioned in [25, 26].

In particular, the axiom system IS(A) from [25, 26] constituted an exception to the Semantic Tableaux version of the Second Incompleteness Theorem with the following three characteristics:

1. Let the atomic symbol $M(x, y, z)$ represent that the multiplicative product of x times y equals z . IS(A) contained all the axioms of Q, except that it did not recognize Multiplication as a total function. This means that it replaced Multiplication's conventional function symbol with the " $M(x, y, z)$ " atomic predicate symbol, and it did not recognize the validity of the axiom " $\forall x \forall y \exists z : M(x, y, z)$ ".
2. IS(A) had a capacity to prove all the Π_1 theorems of Peano Arithmetic when they are encoded in a modified language that replaces the the Multiplication Function symbol with the $M(x, y, z)$ -atomic

predicate symbol. Moreover, its “Group-2 Axioms” assured that each such Peano Arithmetic Π_1 theorem would satisfy Definition 6.1’s requirement for having a “Hyper-Constructive” proof.

3. IS(A) also had a capacity to recognize no Semantic Tableaux proof of $0=1$ from itself existed.

The pleasing aspect of Theorem 6.6 is that it implies that it is impossible to revise IS(A) so that it can recognize Multiplication as a total function while maintaining its features (2) and (3) simultaneously. This is because Feature (2) implies the existence of a Hyper-Constructive Semantic proof of V^+ (simply because V^+ is one of Peano Arithmetic Π_1 theorems). This feature thus assures that Theorem 6.6 will become automatically applicable to every revised form of IS(A) that possibly recognizes Multiplication as a total function. Thus by Theorem 6.6, every such consistent revision of IS(A) will lose its ability to recognize the non-existence of a Semantic Tableaux proof of $0=1$ from itself.

Hence, the preceding paragraph has illustrated a fairly tight match between our new generalizations of the Second Incompleteness Theorem and the prior exceptions to it in [25, 26]. In conjunction, these complementary positive and negative results illustrate how the axiom recognizing Multiplication as a total function is crucial for effectuating the Second Incompleteness Theorem.

7 Conclusion

We have illustrated two different examples of a pair of positive generalizations of the Second Incompleteness Theorem and negative exceptions to it in this article. (The first example was summarized by Remark 5.6, and the second appeared in the closing two paragraphs of Section 6). In both cases, the very act of recognizing Multiplication as a total function had served as the trigger point causing the Semantic Tableaux version of the Second Incompleteness Theorem to become valid. (Moreover, a byproduct of our formalism is that it shows that the Semantic Tableaux version of the Second Incompleteness Theorem generalizes for IS_0 .)

As an open question for future research, we ask what is the simplest and shortest possible Π_1 sentence V such that $Q + V$ would serve as a threshold for the Semantic Tableaux version of the Second Incompleteness (in the sense that the revised V would satisfy the analogs of Theorems 3.4 and 3.5). It is likely that our current sentence V could be shortened considerably because we have strived to shorten the proofs of Theorems 3.4 and 3.5 in this paper, rather than try to find the most abbreviated possible Π_1 sentence V with these two properties.

Finally, we wish to conclude this article on a mildly amusing note. Many readers will smile with amusement when they learn the *true reason* that the Semantic Tableaux version of the Second Incompleteness Theorem undergoes an entire collapse [25, 26] when Multiplication is changed from a total function into a 3-way relation $M(x, y, z)$. It is essentially that Lemma 4.7 and its *seemingly trivial* short 3-paragraph proof would then become no longer valid. Lemma 4.7 was crucial because we needed the x_0 and x_1 substrings from its proof, as an interim step, to establish the more sophisticated Lemma 4.8. Without Lemma 4.7’s seemingly quite innocent and simple interim step, all the other successive stages of our proof will collapse in a one-by-one, step-by-step manner.

Appendix: The Δ_0 Encoding of $\text{Log}(x, u)$'s Graph

Our main goal in this appendix is to substantiate the claims that Item 1 of Lemma 3.1 made about the Δ_0 encoding of $\text{Log}(x, u)$'s graph. In the special degenerate case where $u = 1$, the Δ_0 encoding of $\text{Log}(x, 1)$'s graph already appeared in Benett's dissertation [4]. Our further results will follow **in a straightforward manner** by combining Benett's analysis with the theory of LinH functions (discussed by Hájek, Krajíček, Pudlák and Wrathall.)

Some common definitions of a non-deterministic Oracle Turing machine can be found in [9, 10, 29]. The term **Linear Computational Hierarchy** (abbreviated as “ LinH, ”) represents roughly the linear-time analog of computer science's famous Polynomial Hierarchy. This notation was first introduced by Wrathall [29] in the context of the Rudimentary Formulae from Smullyan's dissertation [19].

In order to summarize Wrathall's definition, let n denote the length of an input string x that a multi-tape Turing Machine will process. The following quite conventional definitions from [9, 29] are employed:

1. **DTIME**($f(n)$) is the set of languages that some deterministic multi-tape Turing Machine can accept within the time bound $f(n)$.
2. **LinTime** is the union of $\text{DTIME}(n)$, $\text{DTIME}(2n)$, $\text{DTIME}(3n)$,
3. **NLinTime**(σ) is similarly the set of languages that are accepted by a non-deterministic *linear time* Turing machine that has access to an oracle σ .
4. Σ_i^{lin} is defined to equal LinTime when $i = 0$, and it will equal NLinTime($\Sigma_{i-1}^{\text{lin}}$) when $i > 0$.
5. **LinH** is defined to be the union of Σ_0^{lin} , Σ_1^{lin} , Σ_2^{lin} ,

Wrathall [29] studied LinH largely because it was Logic's analog of Computer Science's well-known Polynomial-Time Hierarchy. Although her work focussed on Smullyan's Rudimentary formulae, several authors (such as Hájek, Krajíček and Pudlák [9, 10]) have noticed her theorems generalized to Δ_0 formulae. As a result of this combined work, one thus has the following theorem.

Theorem A.1. A language L can be recognized by a LinH decision procedure *if and only if* there exists a Δ_0 formula $\Psi(s)$ where the s satisfying Ψ are precisely the elements of L .

Define the predicate $P(x, u, z)$ to be true exactly when $\text{Log}(x, u) = z$. Also, for any fixed constant k , let $P_k(x, z)$ be an abbreviation for $P(x, k, z)$. In this notation, the main theorem in Benett's dissertation [4] stated $P_1(x, z)$ could be encoded as a Δ_0 formula (free in two variables x and z). For any fixed constant k , this theorem can be trivially generalized to show that $P_k(x, z)$ also has a Δ_0 encoding (by essentially nesting k iterations of the P_1 predicate within each other).

Our object in this Appendix is to further show that $P(x, u, z)$ can be encoded as a Δ_0 formula (*free in three variables x , u and z*). We will do this mostly by applying Theorem A.1 to Benett's Δ_0 formula $P_1(x, z)$. Before doing this, we need one preliminary Claim:

Claim A.2. Let $P^*(x, u, z)$ denote a predicate that is the same as the predicate $P(x, u, z)$ except that it additionally requires that $z \neq 0$. (In other words, this means that $P^*(x, u, z)$ is a formalization of Equation (42) below.)

$$\text{Log}(x, u) = z \wedge z \neq 0 \quad (42)$$

Then there exists a LinH procedure for recognizing the triples (x, u, z) satisfying the predicate $P^*(x, u, z)$.

Proof. One procedure for determining whether or not the triple (x, u, z) satisfies P^* is a simple non-deterministic procedure that seeks to determine if there exists a (non-deterministically generated) finite sequence $s_0, s_1, s_2, \dots, s_u$ where $s_0 = x$, $s_u = z$, and $\text{Log}(s_i, 1) = s_{i+1}$ for each $i < u$. From the combination of Benett's theorem and Theorem A.1, it is immediately apparent (see footnote ⁵) that a LinH procedure can determine the validity of the preceding " $\text{Log}(s_i, 1) = s_{i+1}$ " condition in $O(1 + \text{Log}(s_i))$ time. Thus, the asymptotic time to evaluate that for each $i < u$, that the preceding quoted phrase is satisfied, is indicated by the summand below:

$$\sum_{i=0}^{u-1} 1 + \text{Log}(s_i) \quad (43)$$

In order to complete our proof of Claim A.2, we need to show that the running time indicated by (43)'s summand has an $O(\text{Log}(x))$ magnitude. This is easy to do because Equation (42)'s definition of $P^*(x, u, z)$ indicated $z \neq 0$. In this context, each s_i in the sequence $s_0, s_1, s_2, \dots, s_u$, will satisfy $(1 + \text{Log}(s_i)) \leq \frac{2}{3}(1 + \text{Log}(s_{i-1}))$ (by a trivial argument given in the footnote ⁶). This bound implies that Equation (43)'s summand has an $O\{1 + \text{Log}(s_0)\}$ magnitude. Since by definition $s_0 = x$, we thus obtain that the procedure indicated in the first sentence of Claim A.2's proof is indeed a LinH procedure whose runtime is governed by x 's bit-length. $\square$

Claim A.3. The predicate $P(x, u, z)$ (indicating that $\text{Log}(x, u) = z$) has a Δ_0 encoding.

Proof. The definitions of $P(x, u, z)$ and $P^*(x, u, z)$ easily imply that

$$P(x, u, z) \equiv \{P^*(x, u, z) \wedge z \geq 1\} \vee \{z = 0 \wedge \exists g \leq u - 1 P^*(x, g, 1)\} \quad (44)$$

The key point is that Claim A.2 indicated that the predicates P^* on the right side of Equation (44) can be evaluated by LinH decision procedures. Applying Theorem A.1 to this fact, we conclude that the P^* have Δ_0 encodings. Hence from (44), we conclude that so does $P(x, u, z)$ have a Δ_0 encoding. $\square$

⁵As we already noted, the main theorem from Benett's dissertation [4] stated that the formula $P_1(m, n)$ (indicating that $\text{Log}_2(m) = n$) could be encoded as a Δ_0 formula, free in two variables m and n . Since $m \geq n$ whenever this formula is satisfied, it follows by applying Theorem A.1's if-and-only-if statement (in its left-pointing direction) that a LinH decision procedure can check whether or not an arbitrary ordered pair (m, n) satisfies $P_1(m, n)$ in a time proportional to m 's bit-length (i.e. in $O(1 + \text{Log}(m))$ time). After simply setting $s_i = m$ and $s_{i+1} = n$, it follows that this time is $O(1 + \text{Log}(s_i))$ in a context of a search to determine whether the condition " $\text{Log}(s_i, 1) = s_{i+1}$ " is true.

⁶If the condition " $z \neq 0$ " would have been absent from Equation (42)'s definition of $P^*(x, u, z)$ then the inequality " $(1 + \text{Log}(s_i)) \leq \frac{2}{3}(1 + \text{Log}(s_{i-1}))$ " would NOT necessarily hold because this inequality certainly fails when $s_i = s_{i-1} = 0$. However, this difficulty obviously cannot occur because $P^*(x, u, z)$'s definition precludes it through its inequality " $z \neq 0$ " (In particular, the inequality " $z \neq 0$ " forces $s_i \geq 1$ in a context where $\text{Log}(s_{i-1}) = s_i$.)

The Claim A.3 clearly showed that predicate $P(x, u, z)$ satisfied Item 1 from Lemma 3.1. Moreover using $P(x, u, z)$'s Δ_0 encoding, we can trivially write a Π_1 sentence indicating that $\text{Log}(x, u)$ is a total function. Such a Π_1 sentence would simply be:

$$\{ \forall x \forall u \exists z \leq x P(x, u, z) \} \wedge \{ \forall x \forall u \forall y \forall z P(x, u, y) \wedge P(x, u, z) \Rightarrow y = z \} \quad (45)$$

The formal conjunction of the above equation with its obvious analog for subtraction (where $S(x, u, y)$ replaces $P(x, u, y)$) is a Π_1 sentence satisfying the requirements of Item 2 of Lemma 3.1.

References

- [1] Z. Adamowicz, “On Tableaux consistency in weak theories”, preprints of a manuscript from the Mathematics Institute of the Polish Academy of Sciences (1999). (There were two versions of manuscripts, with the identical title, issued by Adamowicz in 1999. One gave a longer proof of an Incompleteness result for $\text{IS}_0 + \Omega_1$, and the other gave a shorter proof for $\text{IS}_0 + \Omega_2$.) See also the next two references for some slightly different versions of this result involving Herbrand deduction.
- [2] Z. Adamowicz, “Herbrand Consistency and Bounded Arithmetic”, to appear in *Fundamental Mathematica*.
- [3] Z. Adamowicz and P. Zbierski, “On Hebrand consistency in weak theories”, *Archive for Mathematical Logic* 40 (2001) pp. 399-413.
- [4] J. Benett, Ph. D. Dissertation, Princeton University, 1962.
- [5] A. Bezboruah & J. Shepherdson, “Gödel’s Second Incompleteness Theorem for Q”, *Journal of Symbolic Logic* 41 (1976), pp. 503–512.
- [6] S. Buss, Bounded Arithmetic, in *Proof Theory Lecture Notes*, Vol. 3, published by Bibliopolis, Naples, 1986.
- [7] M. Fitting, *First Order Logic and Automated Theorem Proving*, Springer-Verlag, 1990.
- [8] G. Gentzen, “Collected Papers of Gerhard Gentzen”, (English translation by M. Szabo), North Holland (1969).
- [9] P. Hájek & P. Pudlák, *Metamathematics of First Order Arithmetic*, Springer-Verlag, 1991.
- [10] J. Krajíček, *Bounded Propositional Logic and Complexity Theory*, Cambridge University Press 1995.
- [11] G. Kreisel, “A Survey of Proof Theory, Part I” in *Journal of Symbolic Logic* 33 (1968), pp. 321–388, and “Part II” in *Proceedings of Second Scandinavian Logic Symposium* (1971), pp. 109-170, North Holland Press, Amsterdam.
- [12] G. Kreisel & G. Takeuti, “Formally Self-Referential Propositions for Cut-Free Classical Analysis and Related Systems”, *Dissertationes Mathematicae* 118 (1974), pp. 1–55.
- [13] E. Mendelson, *Introduction to Mathematical Logic*, Wadsworth-Brooks-Cole Math Series, 1987. See page 157 for Q’s definition.

- [14] E. Nelson, *Predicative Arithmetic*, Princeton University Press, 1986.
- [15] J. Paris & A. Wilkie, “ Δ_0 Sets and Induction”, *Proceedings of the Jadswin Logic Conference (Poland)*, Leeds University Press (1981), pp. 237–248.
- [16] P. Pudlák, “Cuts Consistency Statements and Interpretations”, *Journal of Symbolic Logic* 50 (1985), pp. 423–442.
- [17] P. Pudlák, “On the Lengths of Proofs of Consistency”, in *Collegium Logicum: Annals of the Kurt Gödel Society Volume 2*, pp. 65–86, published (1996) by Springer-Wien-NewYork in cooperation with Technische Universität Wien.
- [18] R. Robinson, “An Essentially Undecidable Axiom System”, *Proceedings of 1950 International Congress on Mathematics*, pp. 729-730, and/or page 157 of ref. [13].
- [19] R. Smullyan, *The Theory of Formal Systems*, *Annals of Math Studies (Volume 47)* Princeton Univ Press, 1961.
- [20] R. Smullyan, *First Order Logic*, Springer-Verlag, 1968.
- [21] R. Solovay, Private Communications (1994) about Pudlák’s main theorem from [16]. See Appendix A of [26] for a 4-page summary of Solovay’s idea.
- [22] G. Takeuti, “On a Generalized Logical Calculus”, *Japan Journal on Mathematics* 23 (1953), pp. 39–96.
- [23] G. Takeuti, *Proof Theory*, *Studies in Logic Volume 81*, North Holland, 1987.
- [24] A. Wilkie & J. Paris, “On the Scheme of Induction for Bounded Arithmetic”, *Annals on Pure and Applied Logic* 35 (1987), pp. 261–302.
- [25] D. Willard, “Self-Verifying Axiom Systems”, *Third Kurt Gödel Symp (1993)*, pp. 325–336, Springer-Verlag LNCS#713. See also [26] for more details.
- [26] D. Willard, “Self-Verifying Systems, the Incompleteness Theorem & Tangibility Reflection Principle”, to appear in the *Journal of Symbolic Logic*.
- [27] D. Willard, “Self-Reflection Principles and NP-Hardness”, *Dimacs Series in Discrete Mathematics & Theoretical Computer Science # 39 (1997)*, pp. 297–320 (AMS Press).
- [28] D. Willard, “The Tangibility Reflection Principle”, *Fifth Kurt Gödel Colloquium (1997)*, Springer-Verlag LNCS#1289, pp. 319–334.
- [29] C. Wrathall, “Rudimentary Predicates and Relative Computation”, *Siam Journal on Computing* 7 (1978), pp. 194–209.